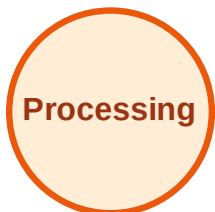


# BFS Traversal

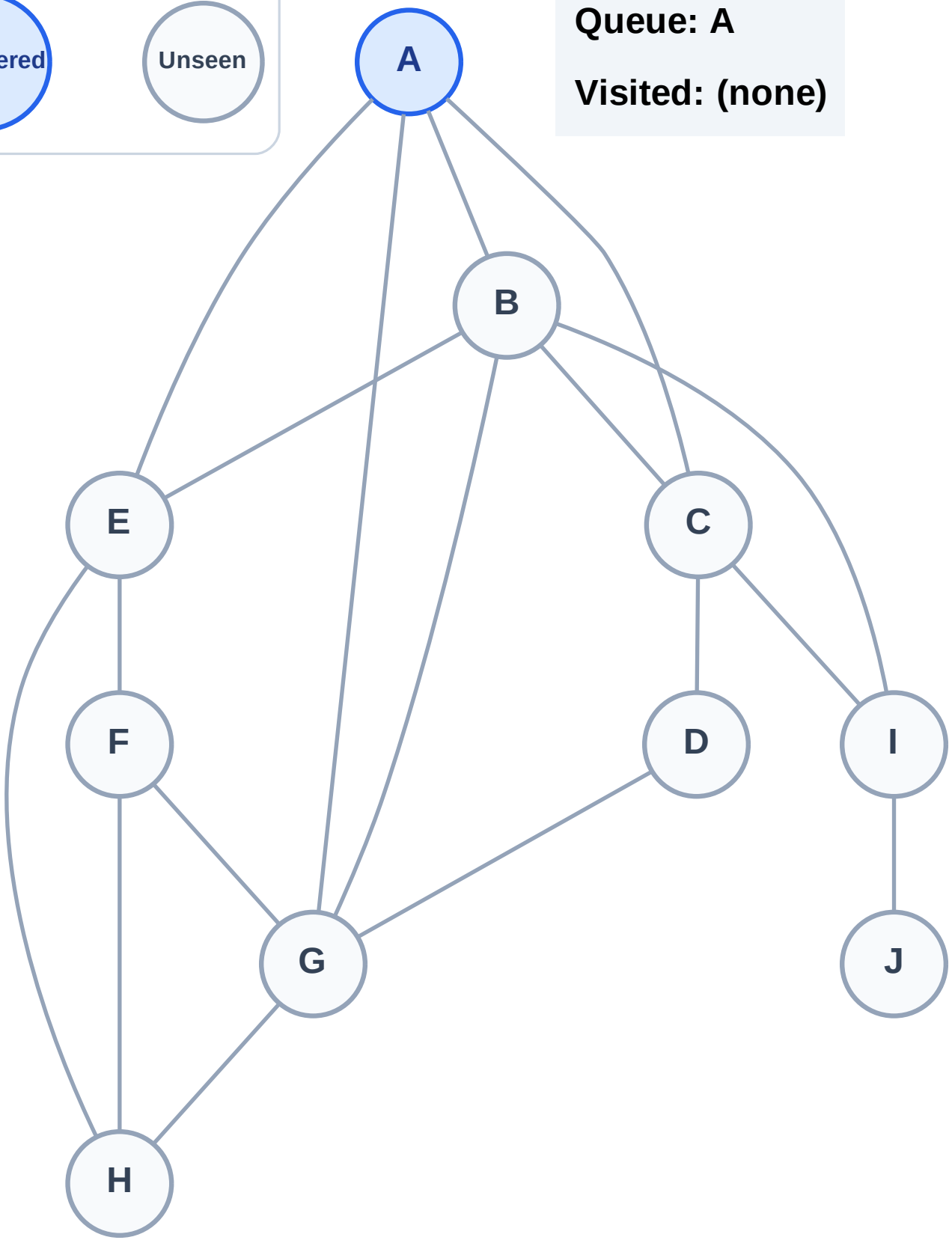
Step 1: Start at A

## Legend



Queue: A

Visited: (none)



# BFS Traversal

Step 2: Process node A

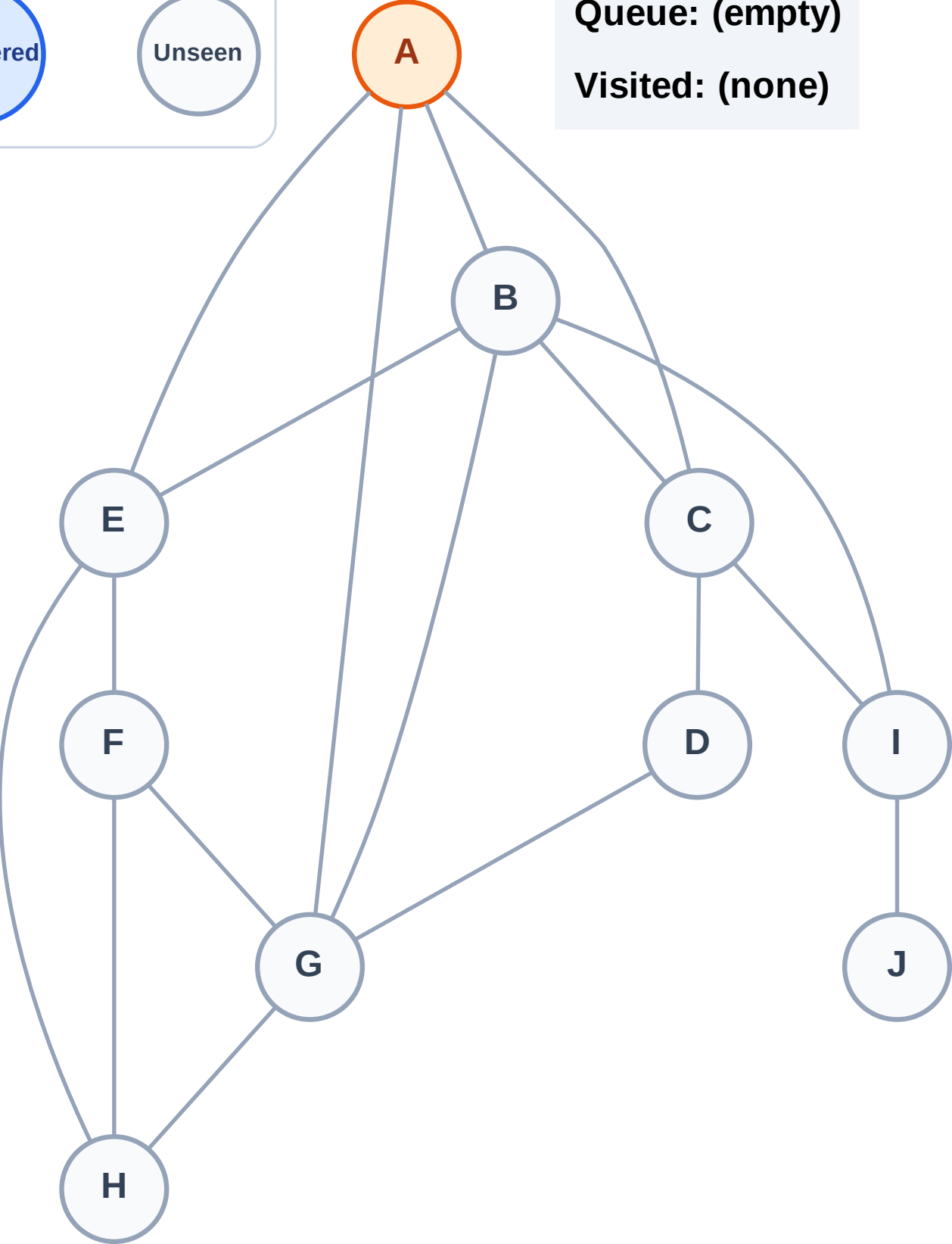
Complete

Processing

Discovered

Unseen

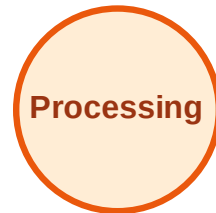
Queue: (empty)  
Visited: (none)



# BFS Traversal

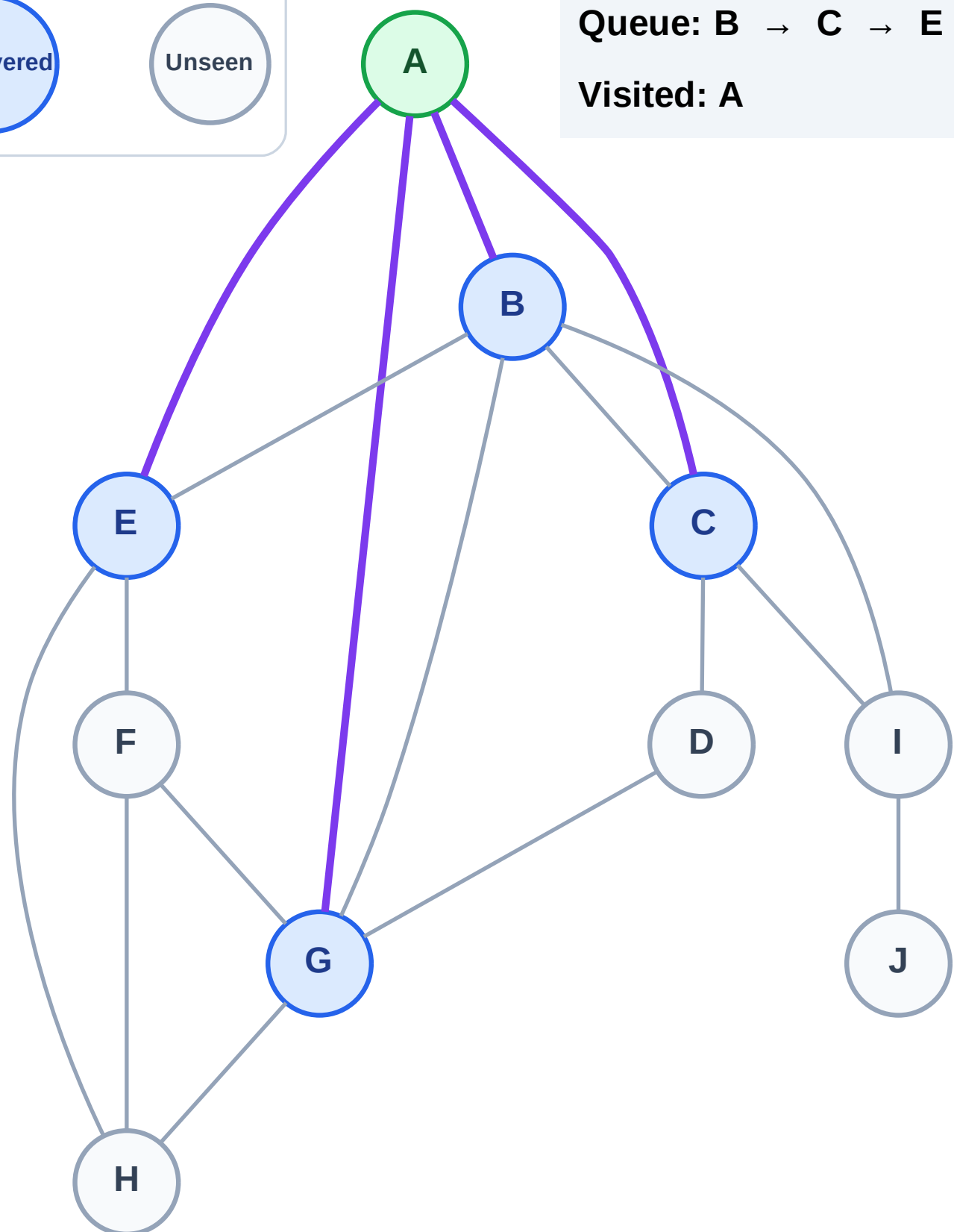
Step 3: Discover B, C, E, G from A

## Legend



Queue: B → C → E → G

Visited: A



# BFS Traversal

Step 4: Process node B

Legend

Complete

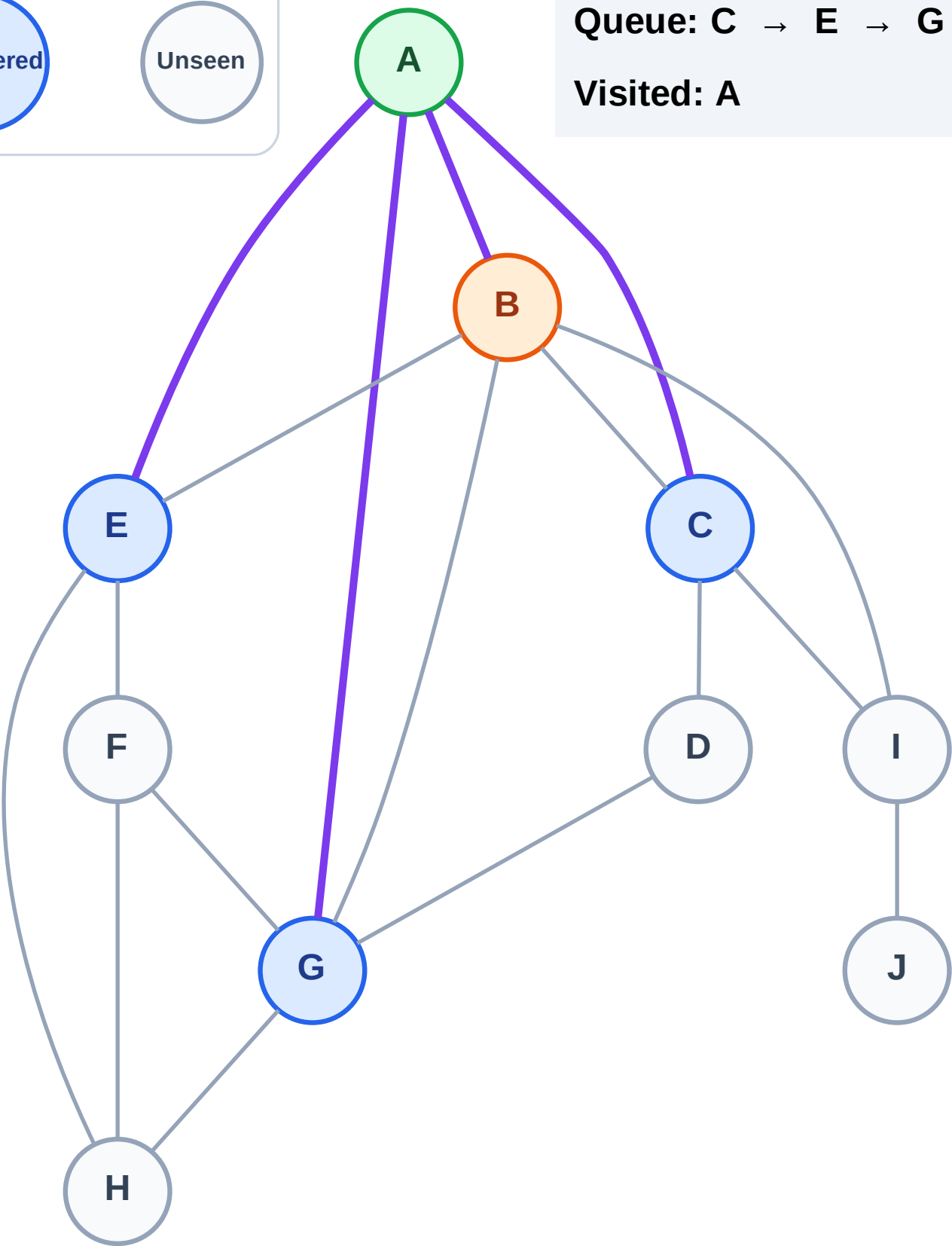
Processing

Discovered

Unseen

Queue: C → E → G

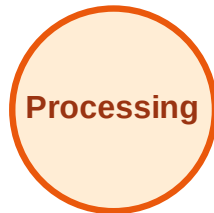
Visited: A



# BFS Traversal

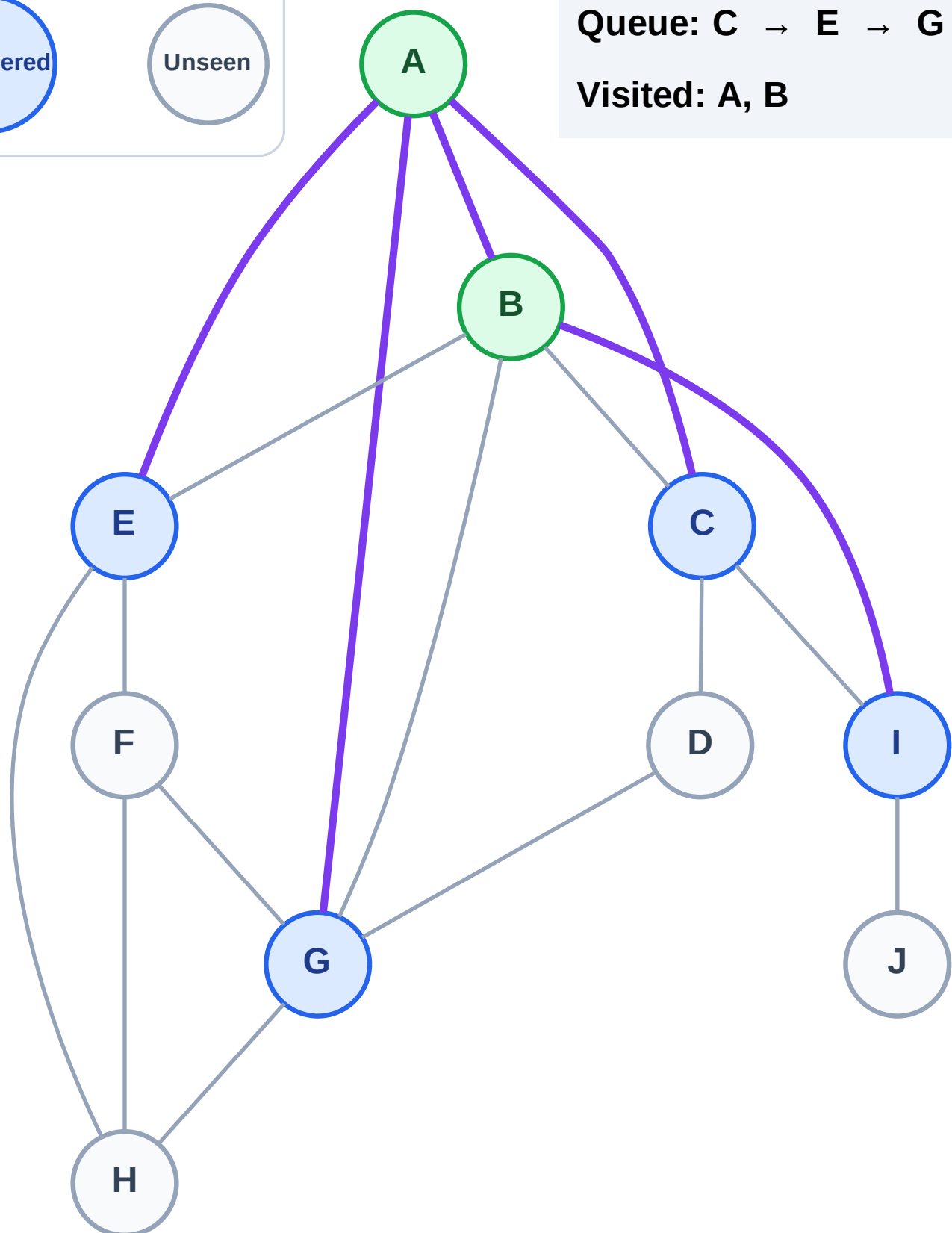
Step 5: Discover I from B

## Legend



Queue: C → E → G → I

Visited: A, B



# BFS Traversal

Step 6: Process node C

Legend

Complete

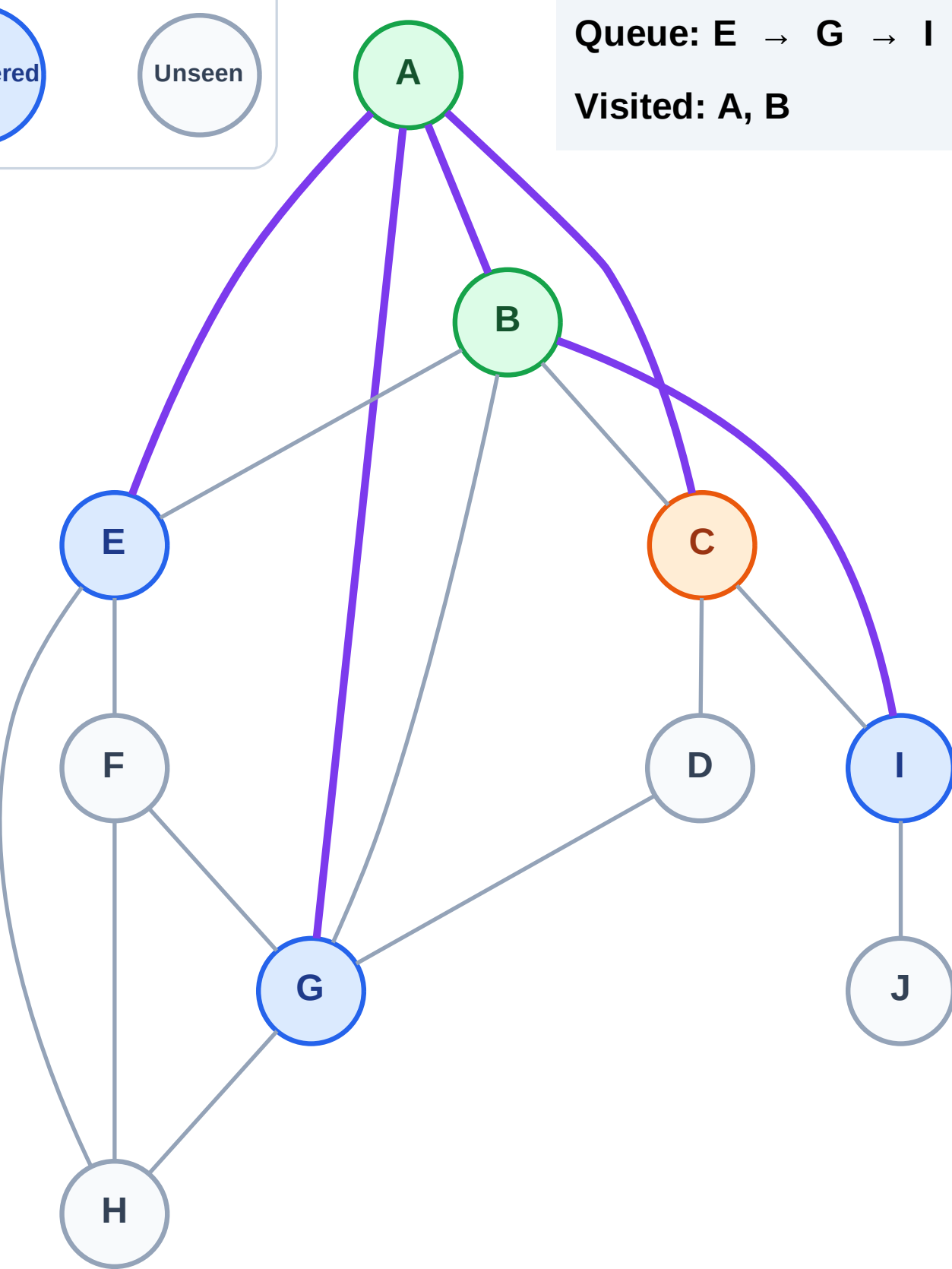
Processing

Discovered

Unseen

Queue: E → G → I

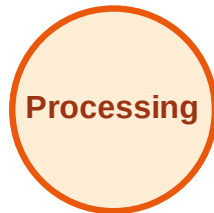
Visited: A, B



# BFS Traversal

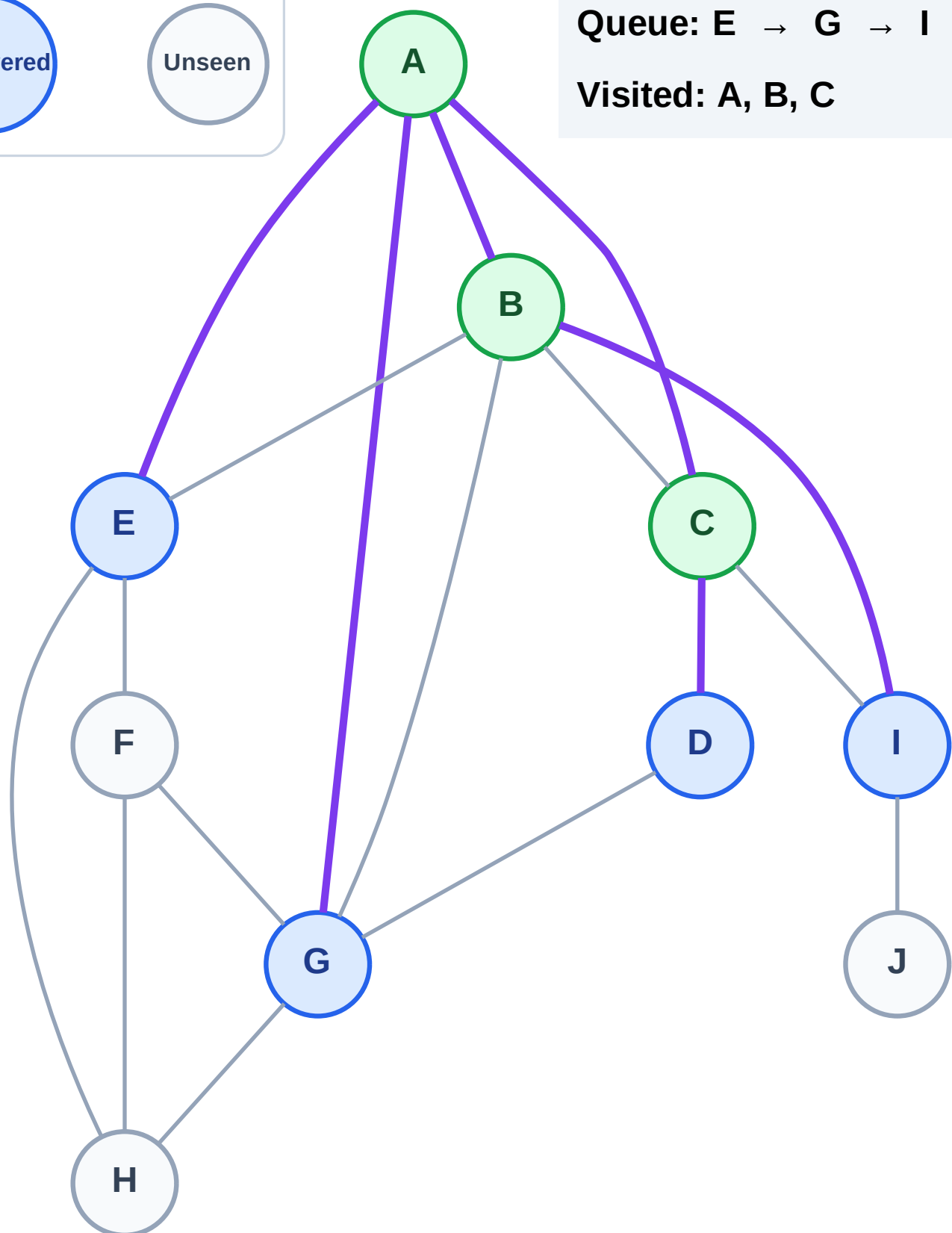
Step 7: Discover D from C

## Legend



Queue: E → G → I → D

Visited: A, B, C



# BFS Traversal

Step 8: Process node E

Legend

Complete

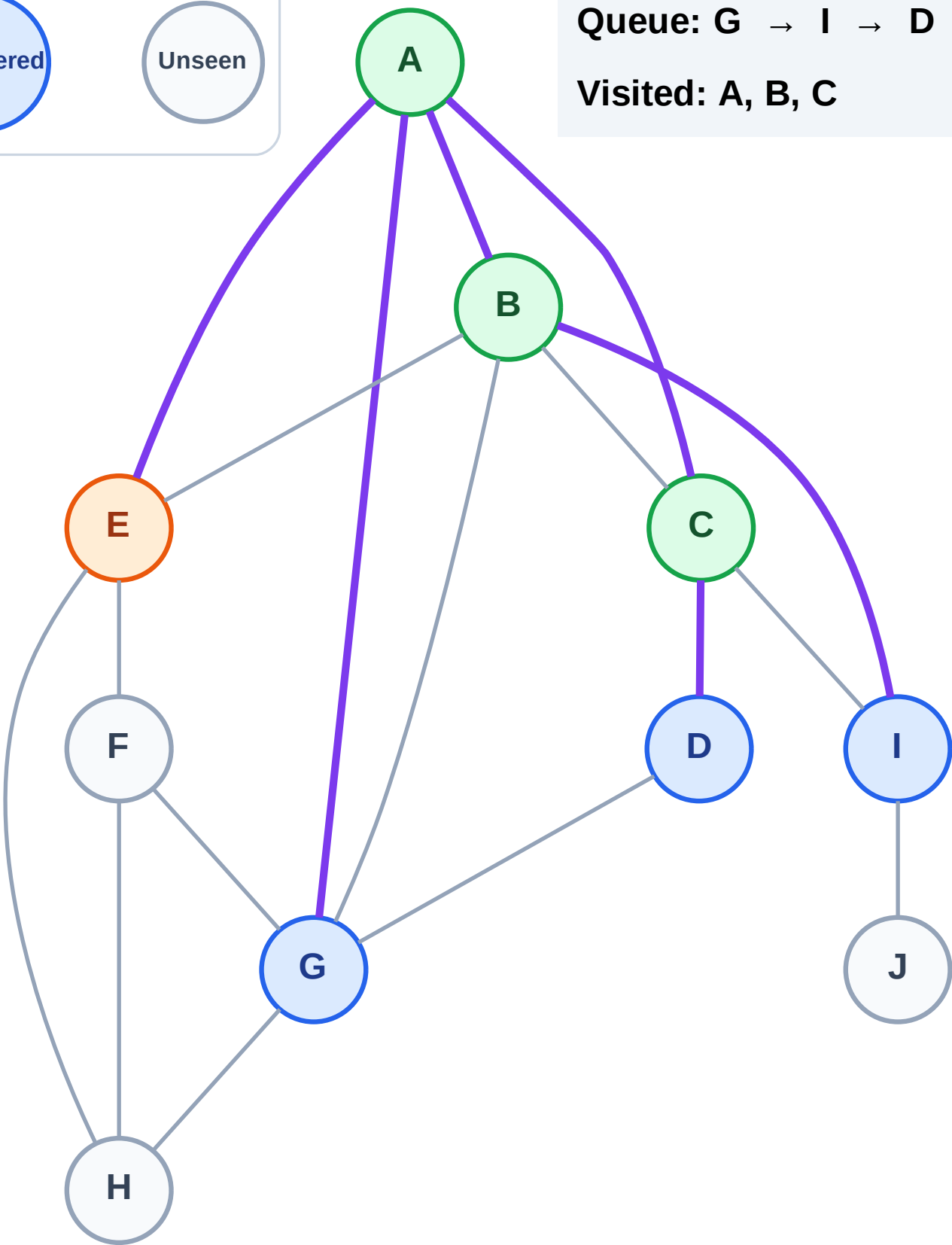
Processing

Discovered

Unseen

Queue: G → I → D

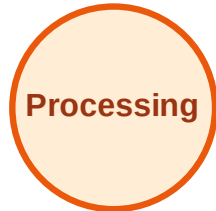
Visited: A, B, C



# BFS Traversal

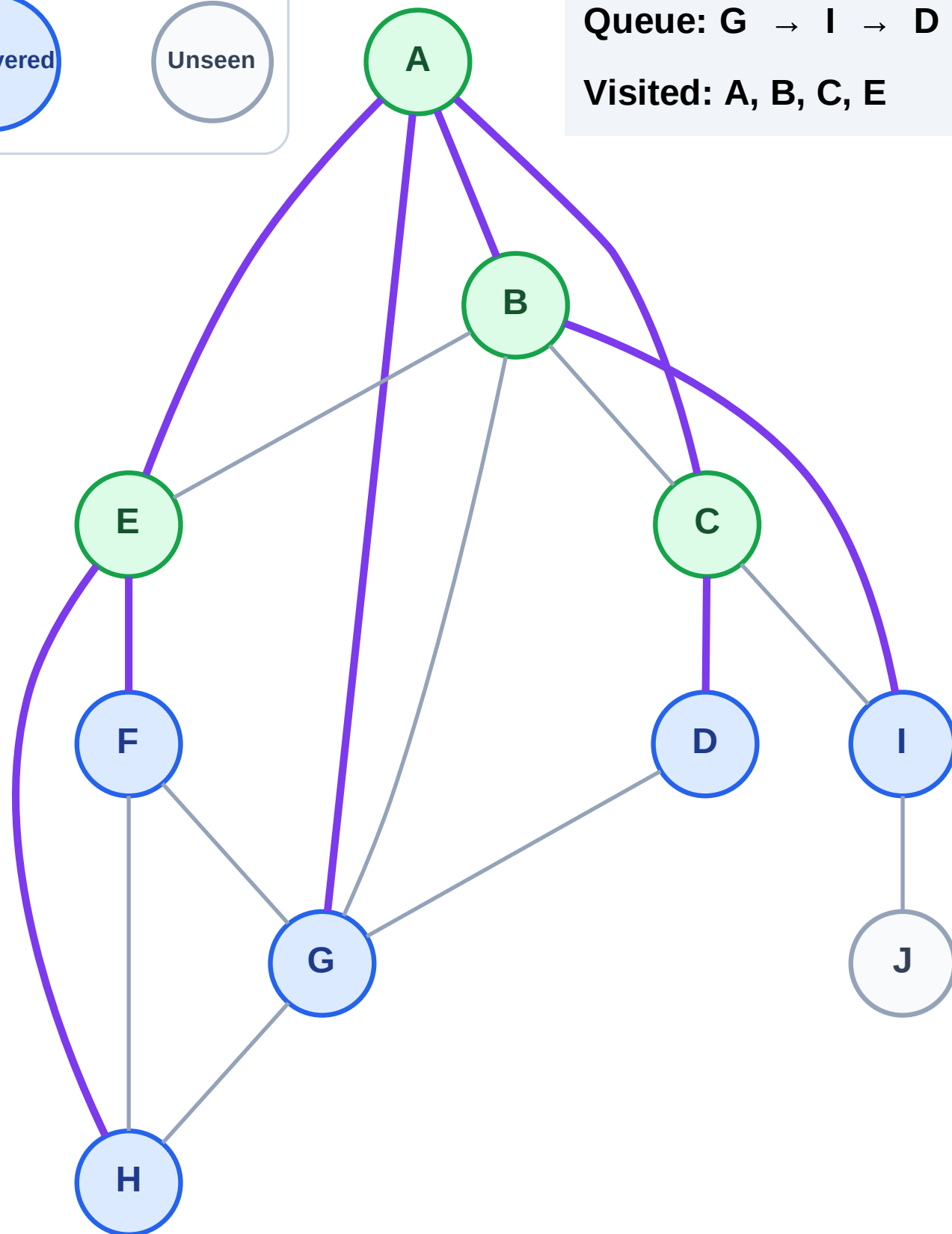
Step 9: Discover F, H from E

## Legend



Queue: G → I → D → F → H

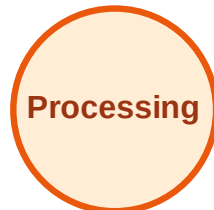
Visited: A, B, C, E



# BFS Traversal

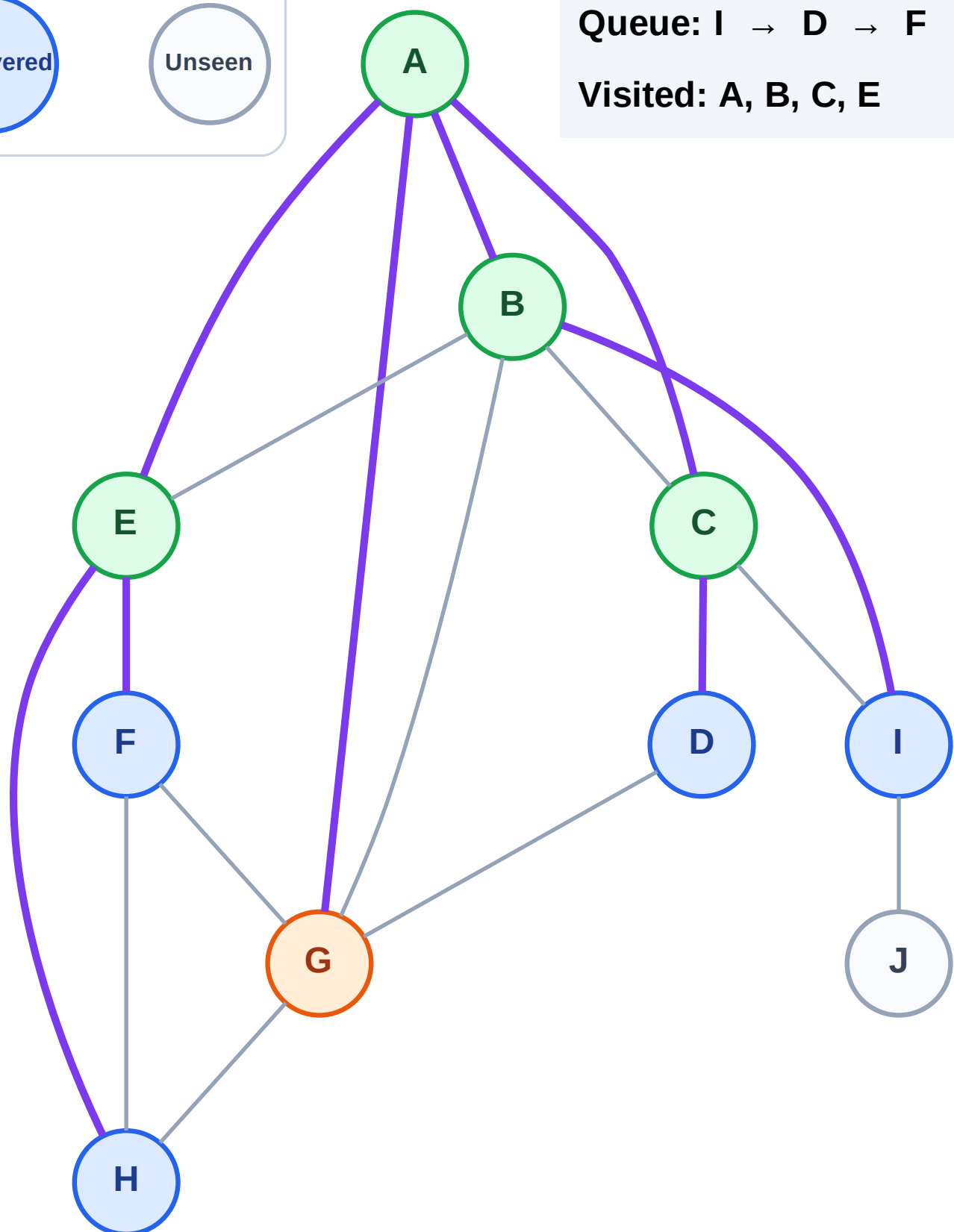
Step 10: Process node G

## Legend



Queue: I → D → F → H

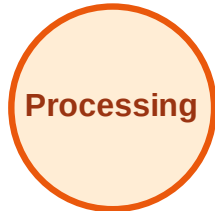
Visited: A, B, C, E



# BFS Traversal

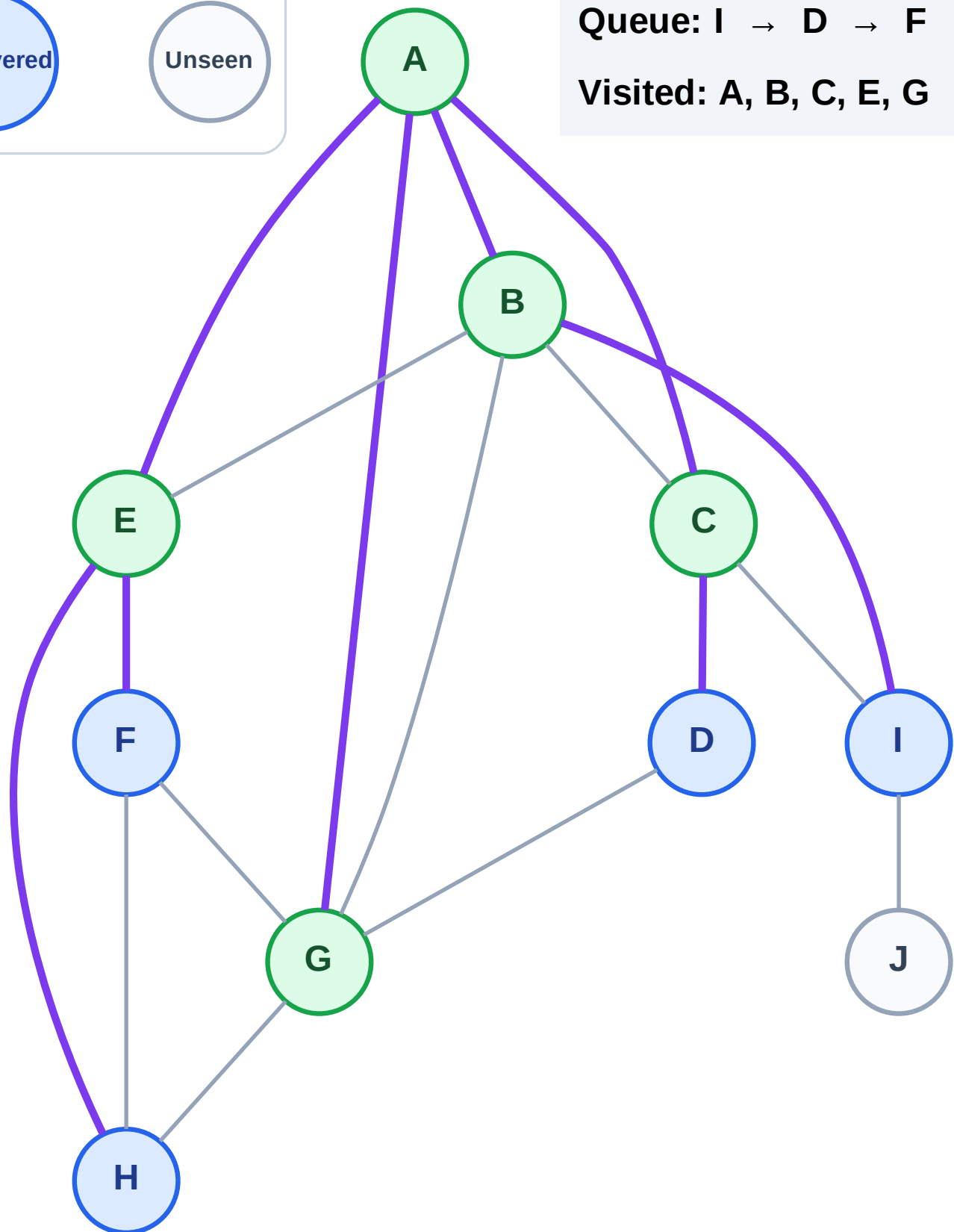
Step 11: No new neighbors from G

## Legend



Queue: I → D → F → H

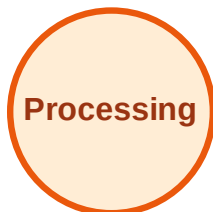
Visited: A, B, C, E, G



# BFS Traversal

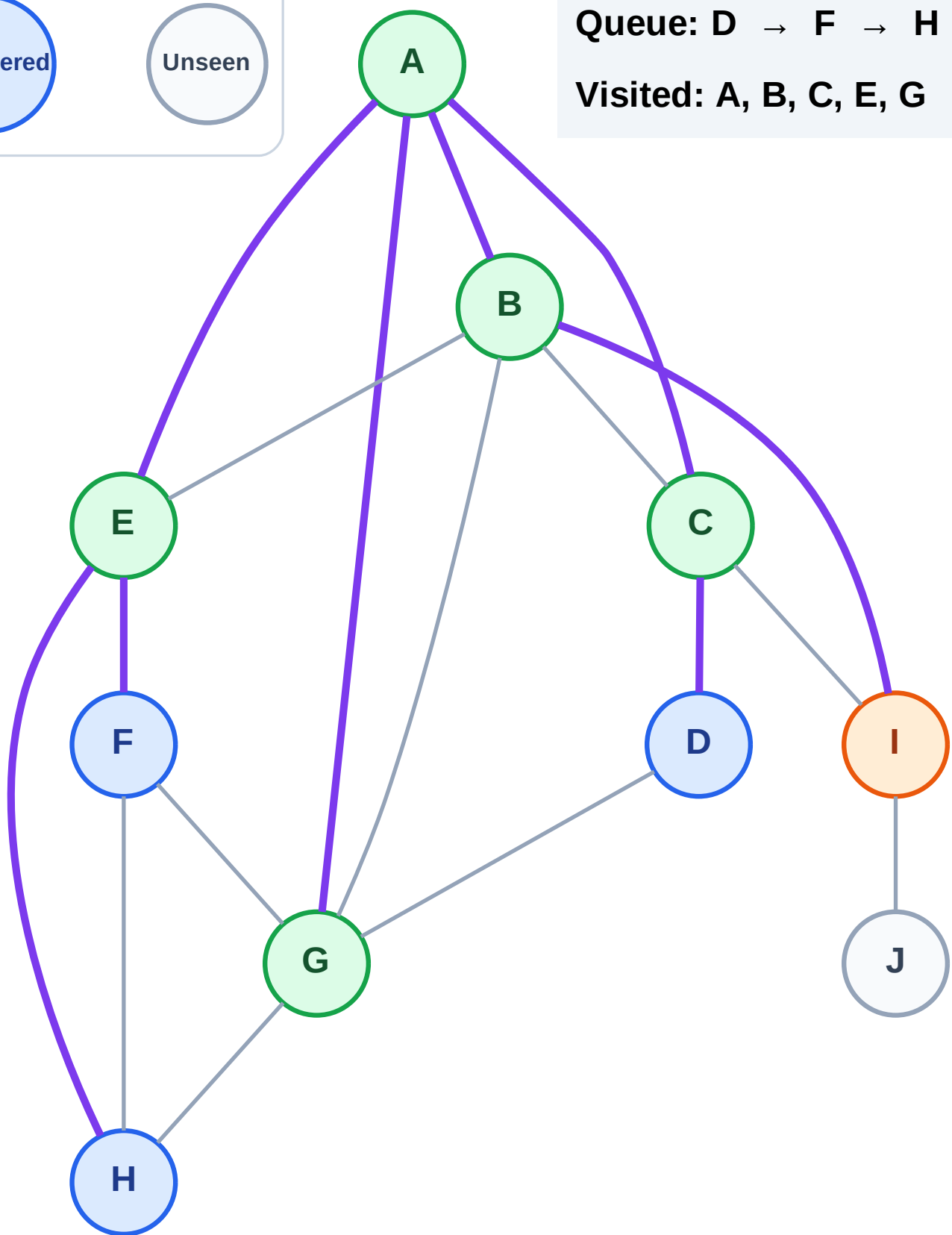
Step 12: Process node I

## Legend



Queue: D → F → H

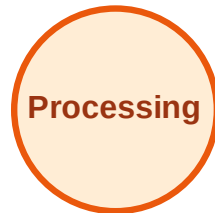
Visited: A, B, C, E, G



# BFS Traversal

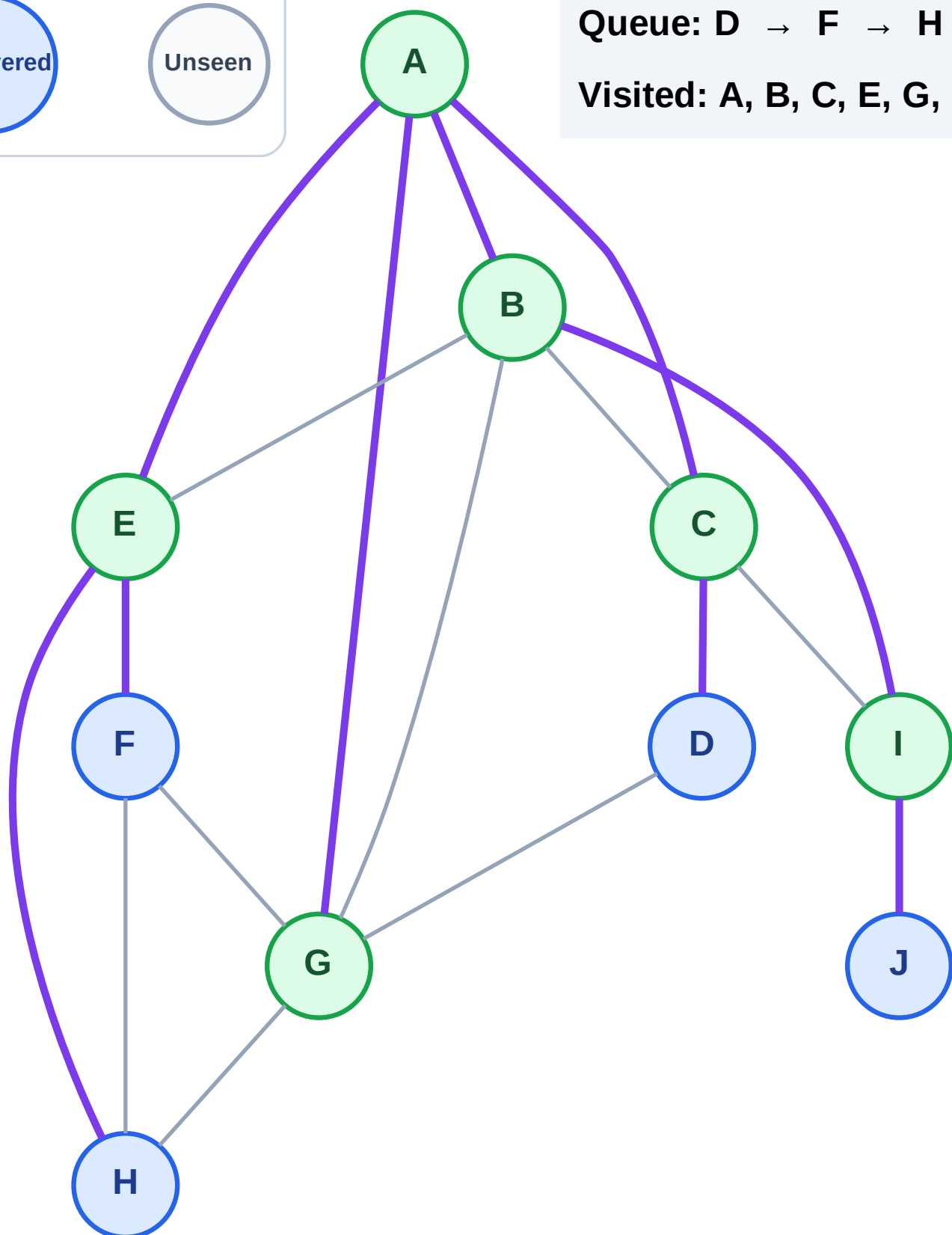
Step 13: Discover J from I

## Legend



Queue: D → F → H → J

Visited: A, B, C, E, G, I



# BFS Traversal

Step 14: Process node D

Legend

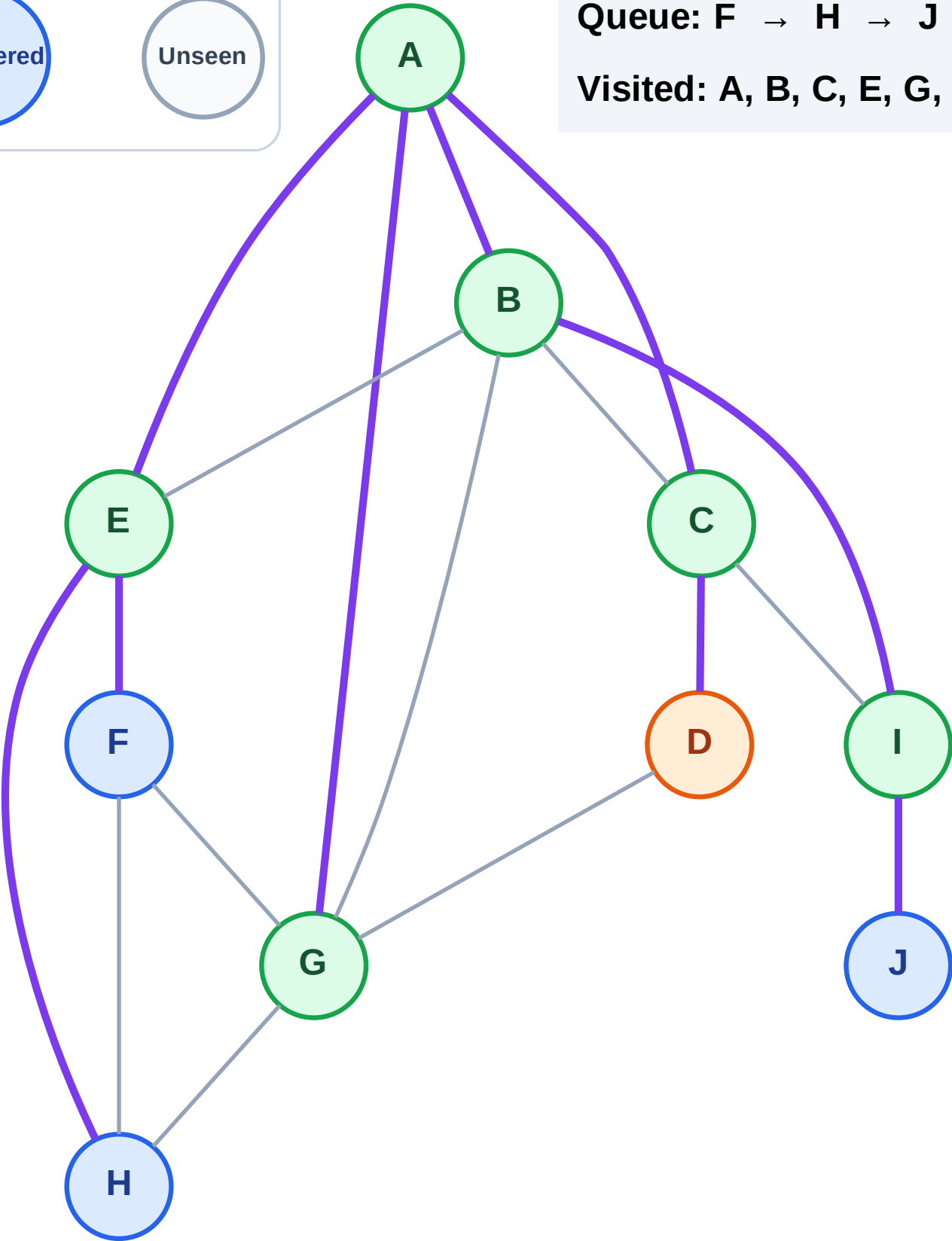
Complete

Processing

Discovered

Unseen

Queue: F → H → J  
Visited: A, B, C, E, G, I



# BFS Traversal

Step 15: No new neighbors from D

Legend

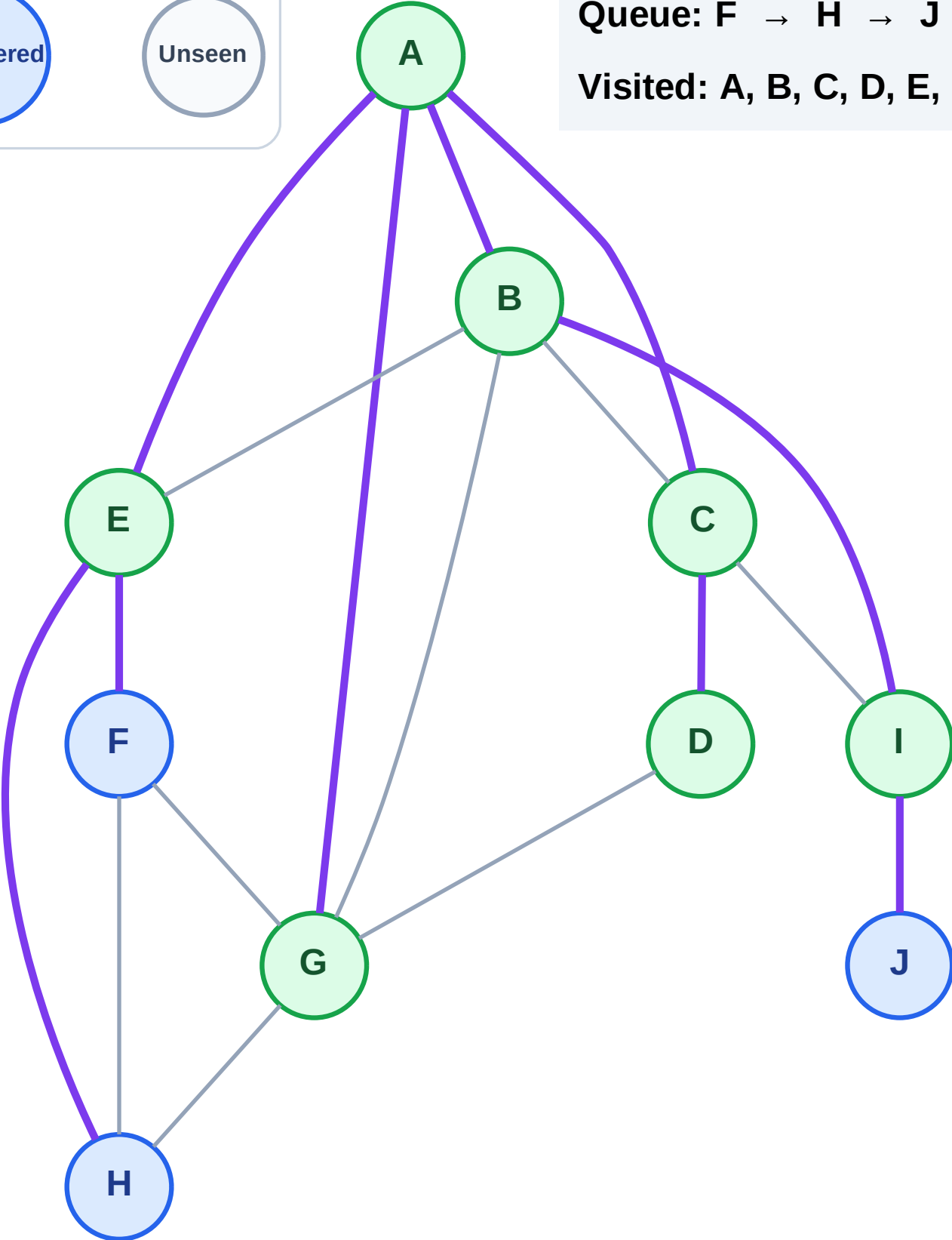
Complete

Processing

Discovered

Unseen

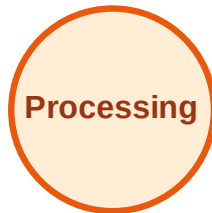
Queue: F → H → J  
Visited: A, B, C, D, E, G, I



# BFS Traversal

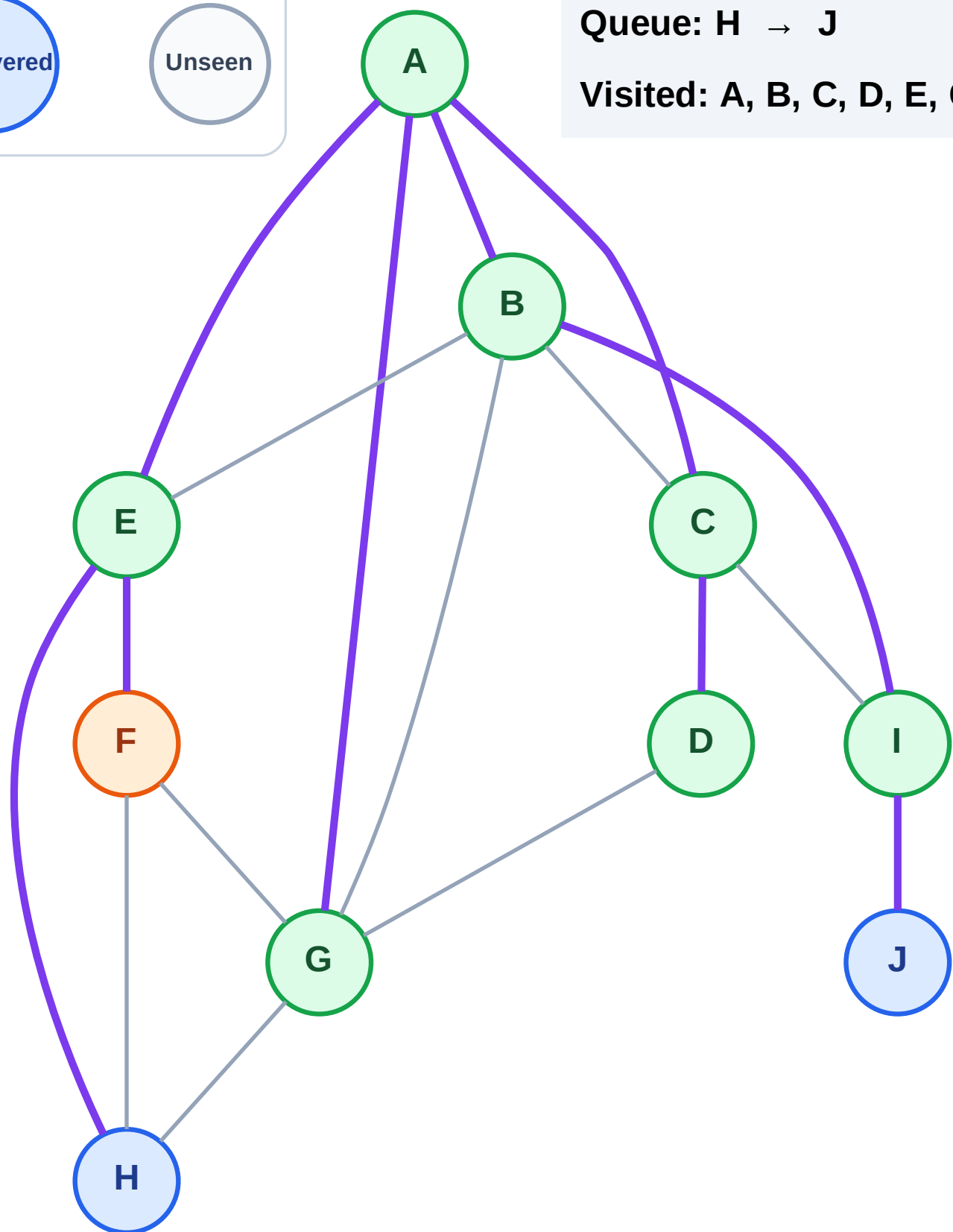
Step 16: Process node F

## Legend



Queue: H → J

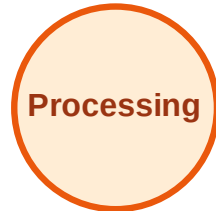
Visited: A, B, C, D, E, G, I



# BFS Traversal

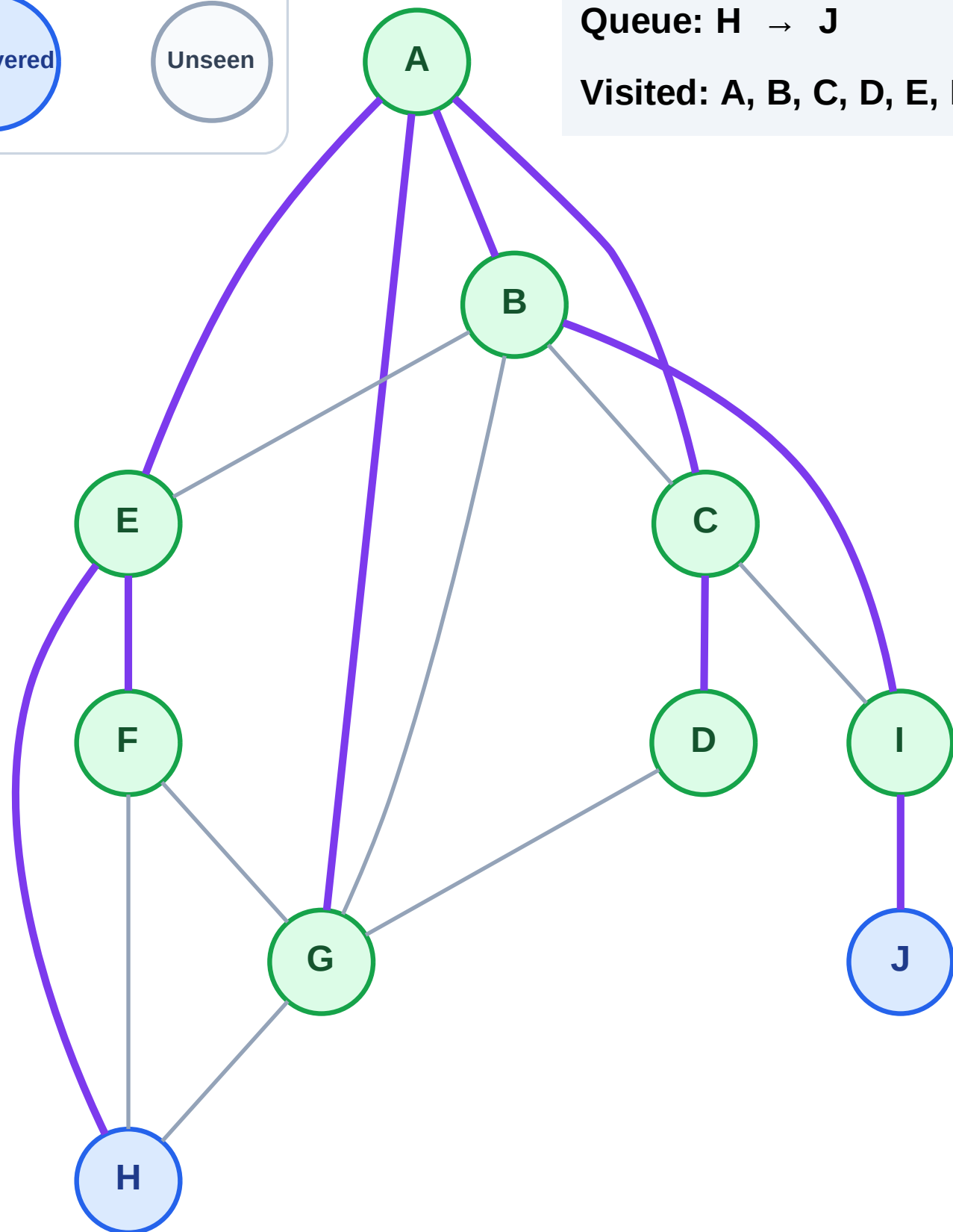
Step 17: No new neighbors from F

## Legend



Queue: H → J

Visited: A, B, C, D, E, F, G, I



# BFS Traversal

Step 18: Process node H

Legend

Complete

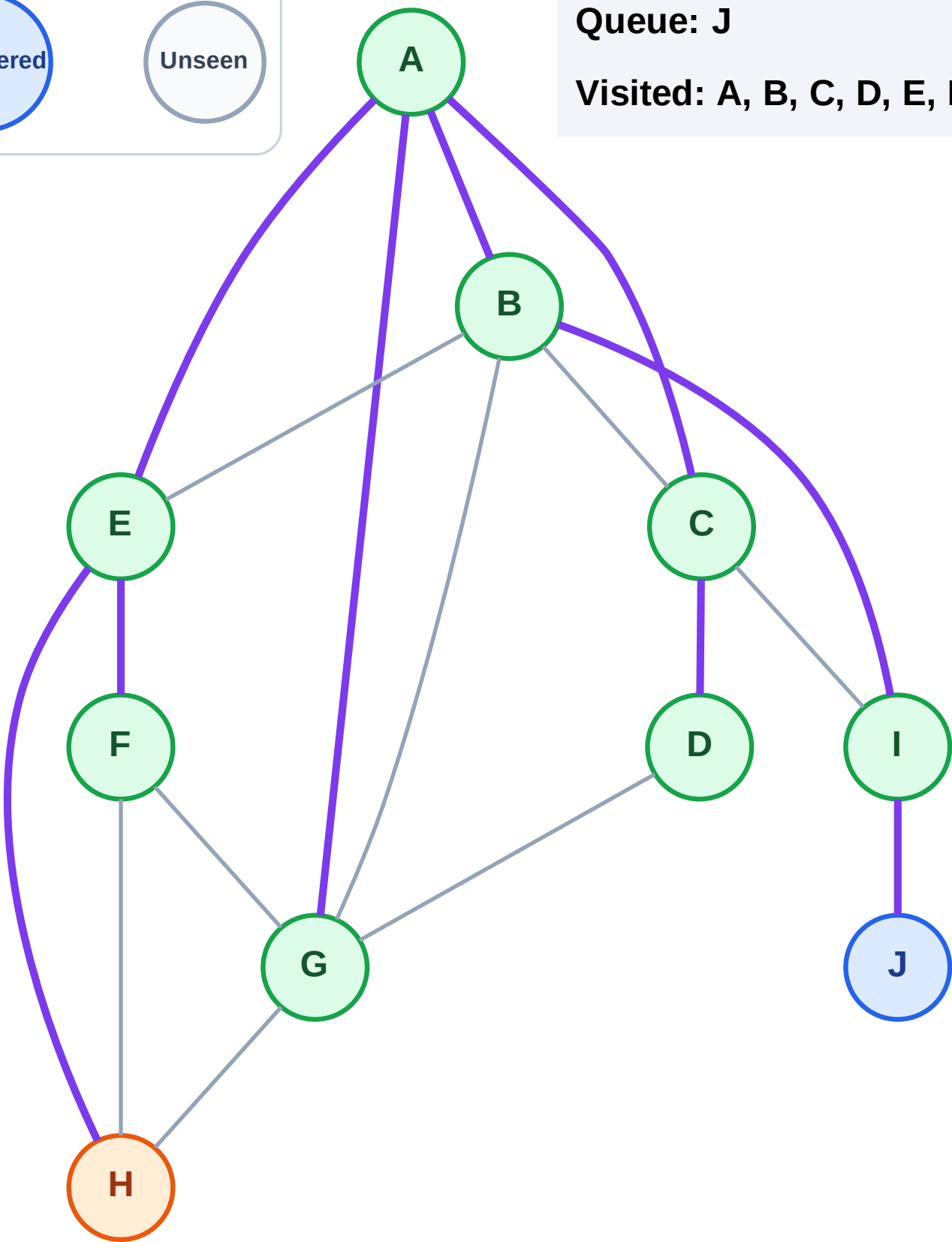
Processing

Discovered

Unseen

Queue: J

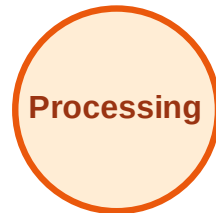
Visited: A, B, C, D, E, F, G, I



# BFS Traversal

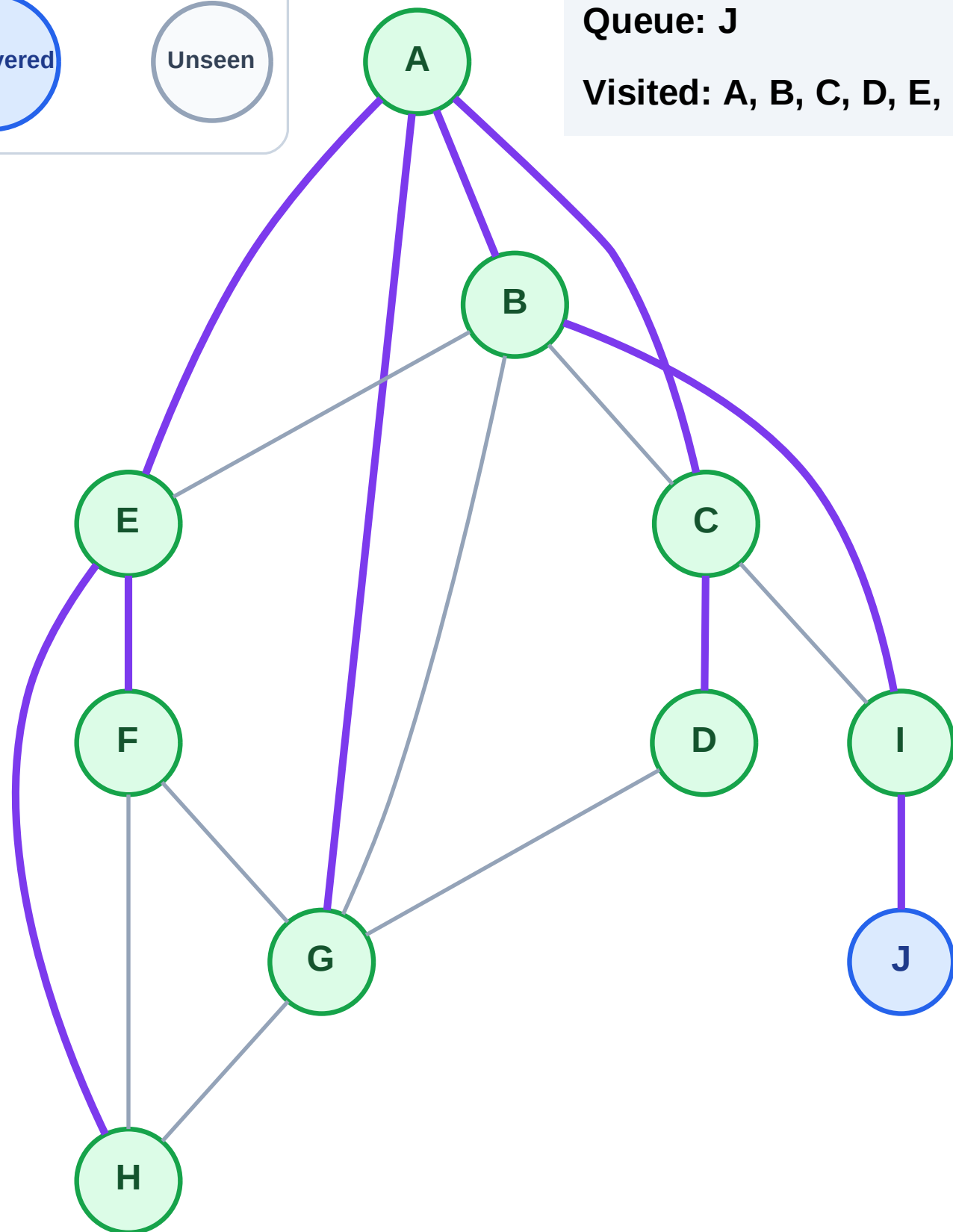
Step 19: No new neighbors from H

## Legend



Queue: J

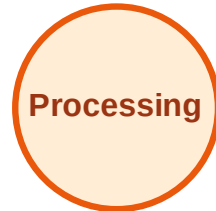
Visited: A, B, C, D, E, F, G, H, I



# BFS Traversal

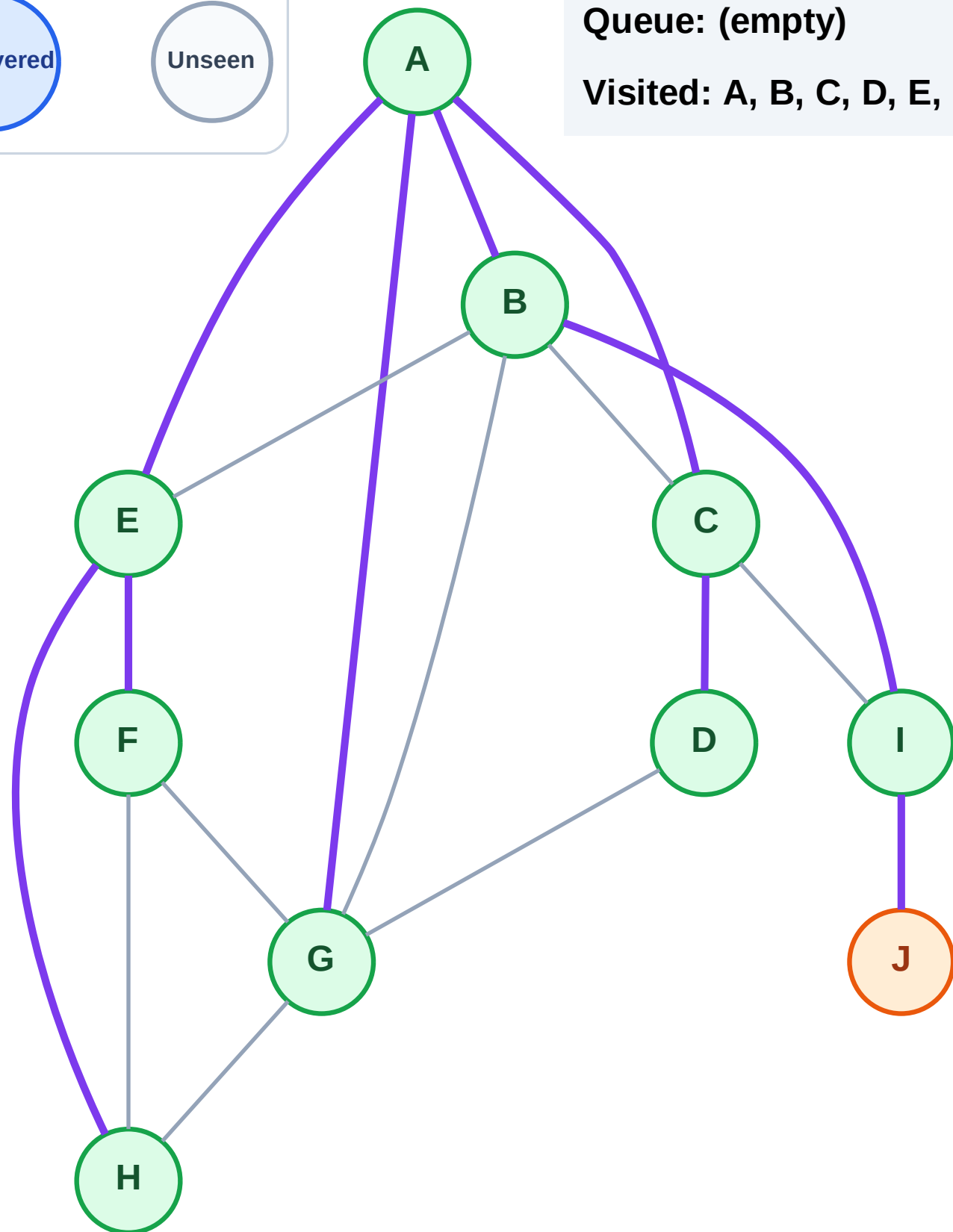
Step 20: Process node J

## Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I



# BFS Traversal

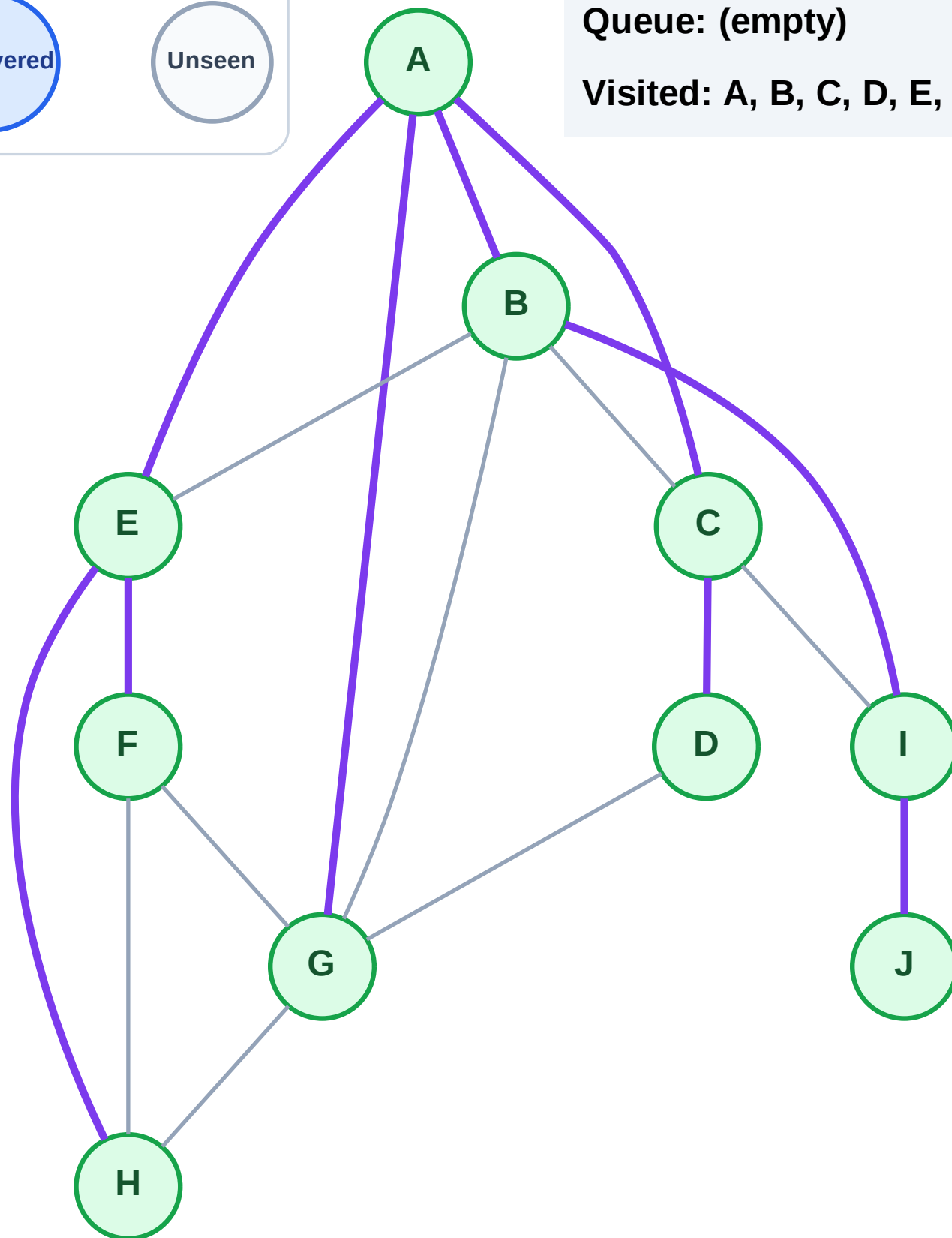
Step 21: No new neighbors from J

## Legend



Queue: (empty)

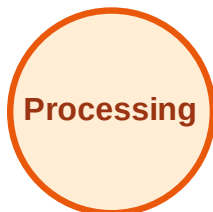
Visited: A, B, C, D, E, F, G, H, I, J



# DFS Traversal

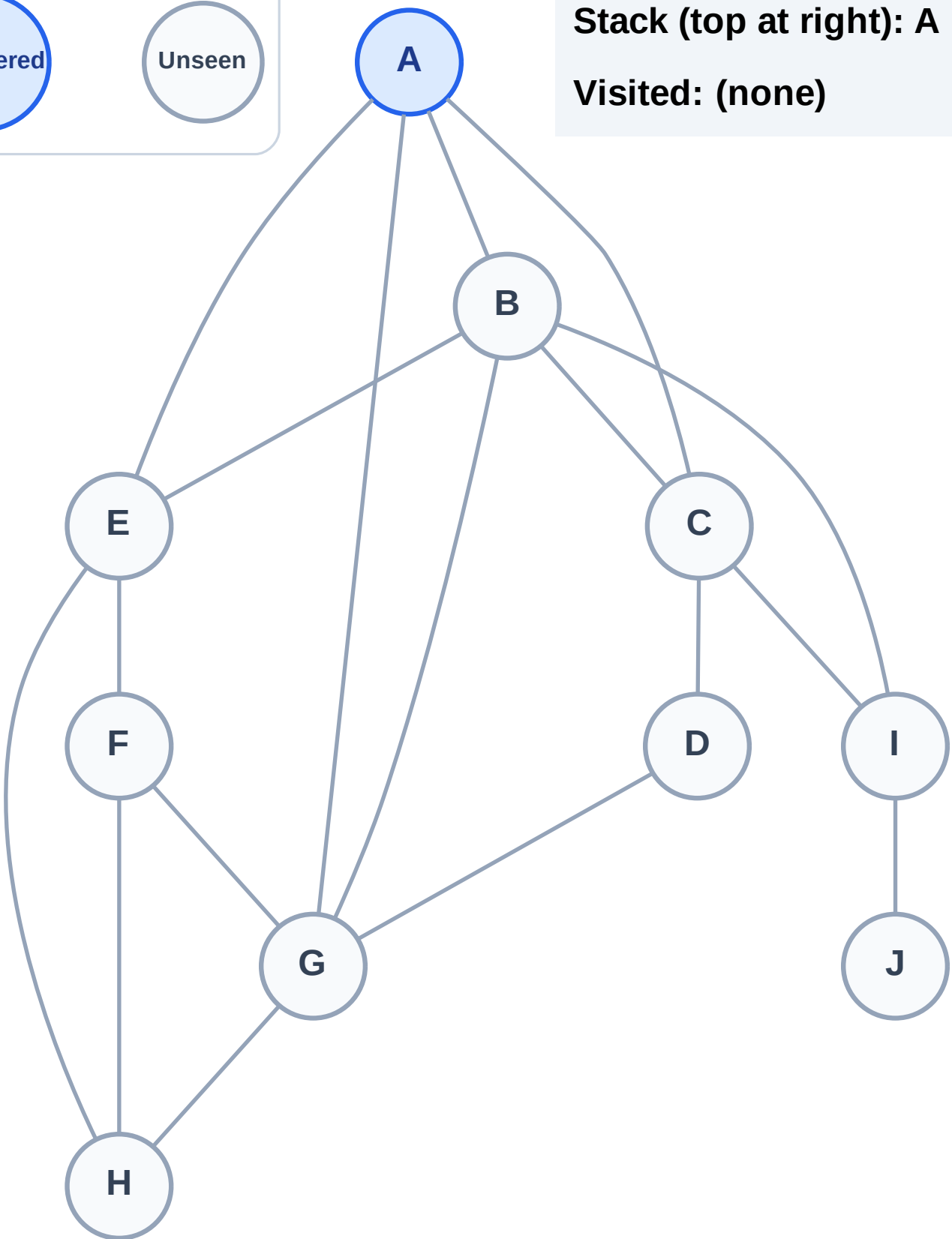
Step 1: Start at A

## Legend



Stack (top at right): A

Visited: (none)



# DFS Traversal

Step 2: Process node A

Legend

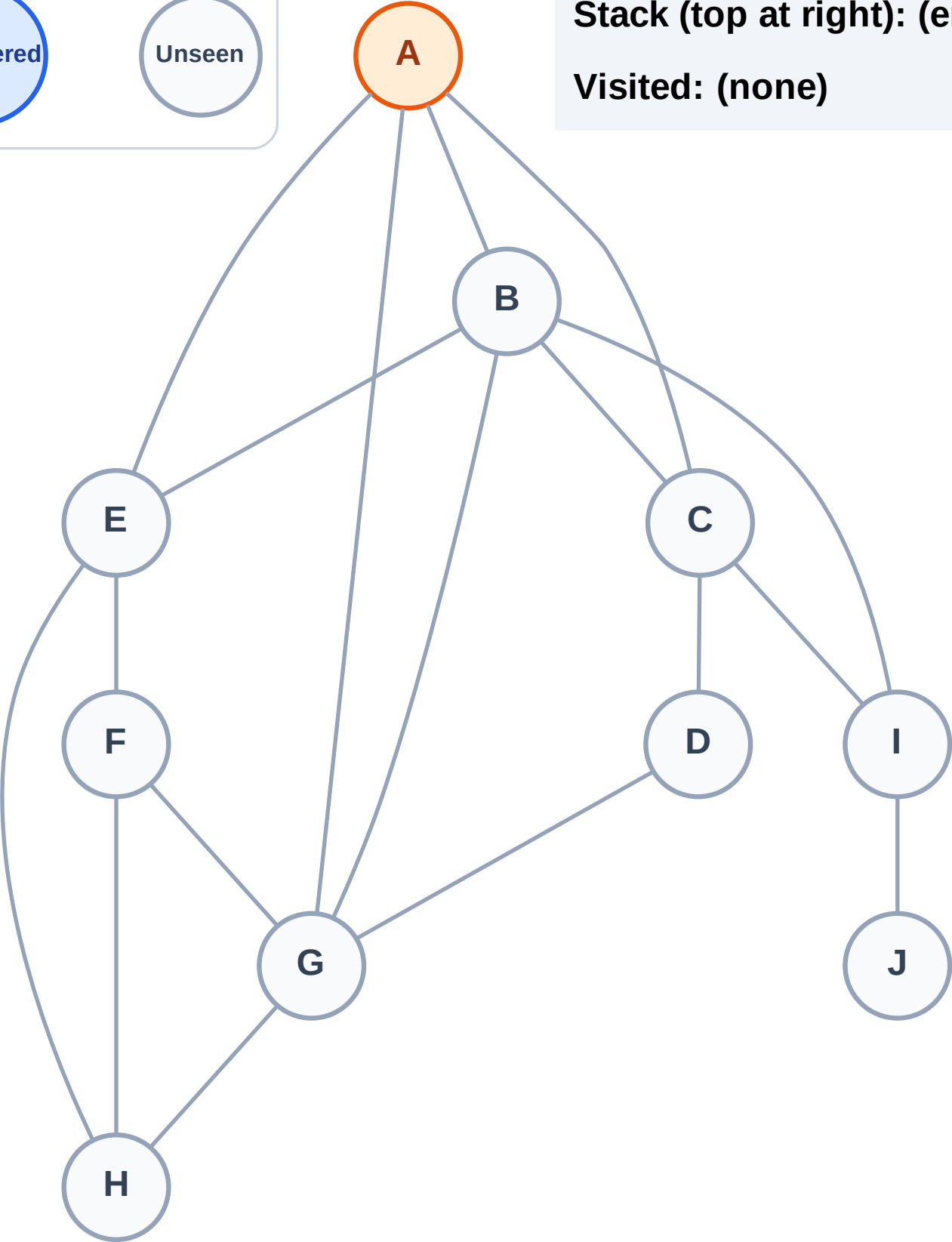
Complete

Processing

Discovered

Unseen

Stack (top at right): (empty)  
Visited: (none)



# DFS Traversal

Step 3: Discover G, E, C, B from A

Legend

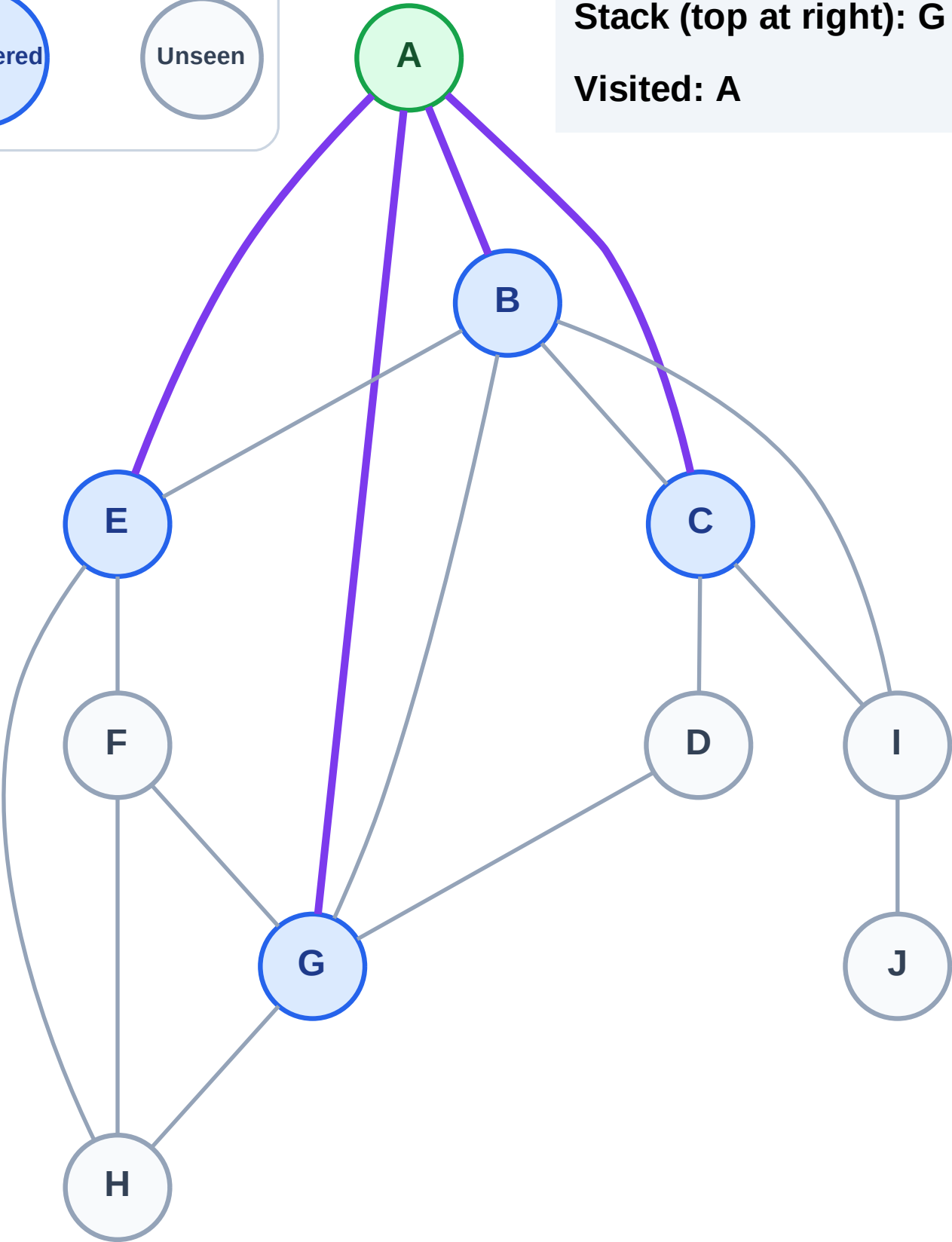
Complete

Processing

Discovered

Unseen

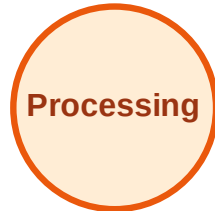
Stack (top at right): G | E | C | B  
Visited: A



# DFS Traversal

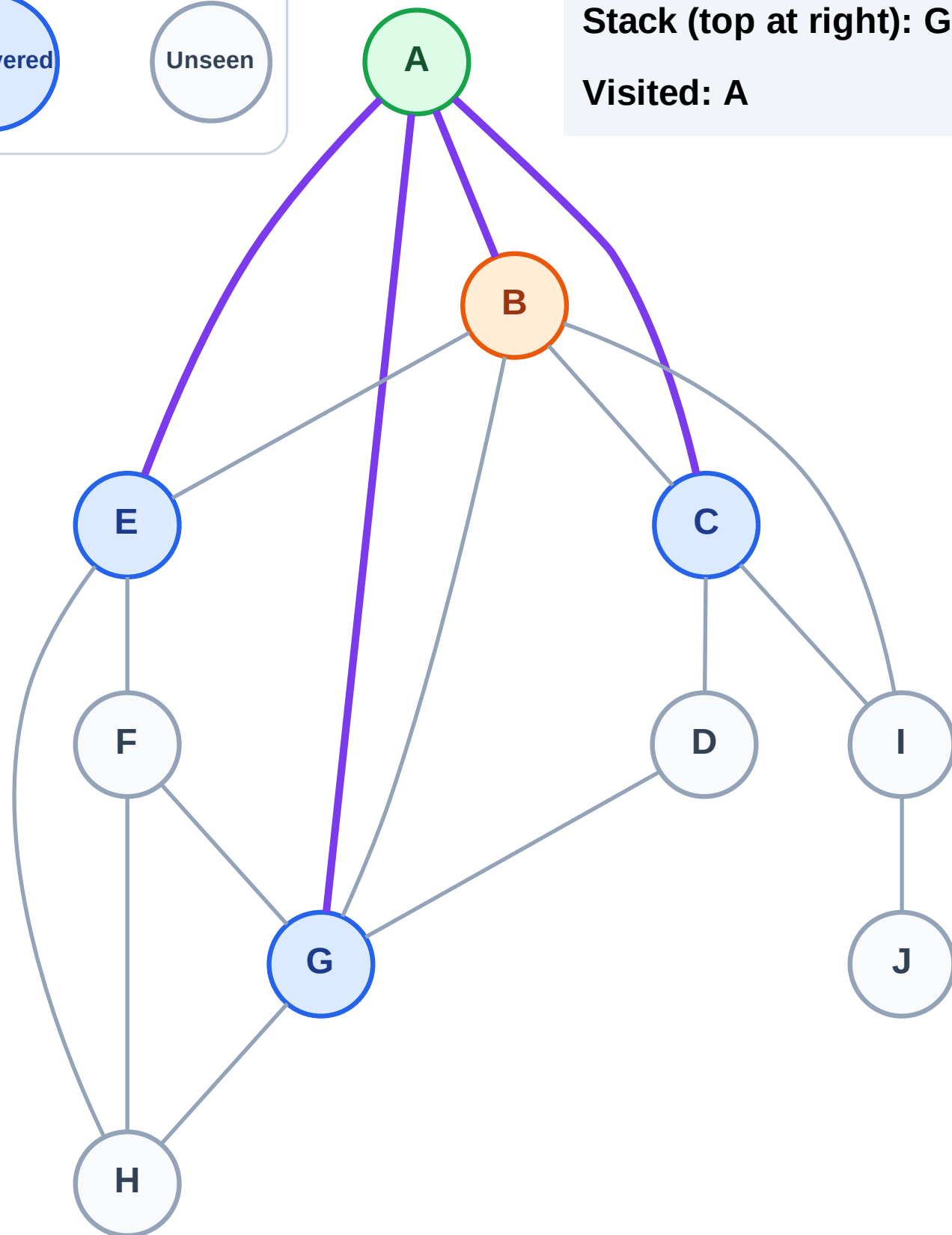
Step 4: Process node B

## Legend



Stack (top at right): G | E | C

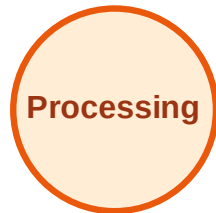
Visited: A



# DFS Traversal

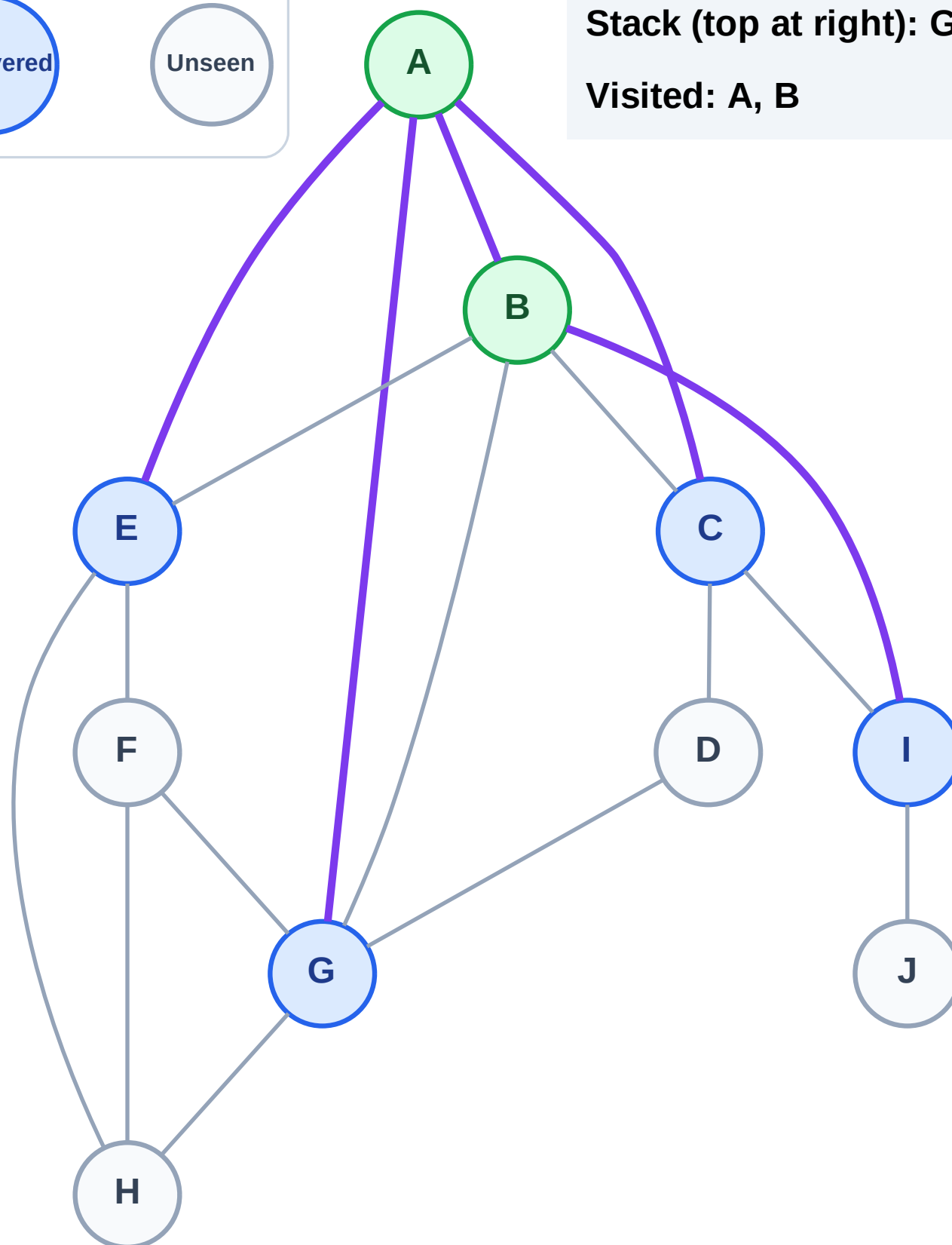
Step 5: Discover I from B

## Legend



Stack (top at right): G | E | C | I

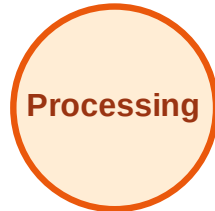
Visited: A, B



# DFS Traversal

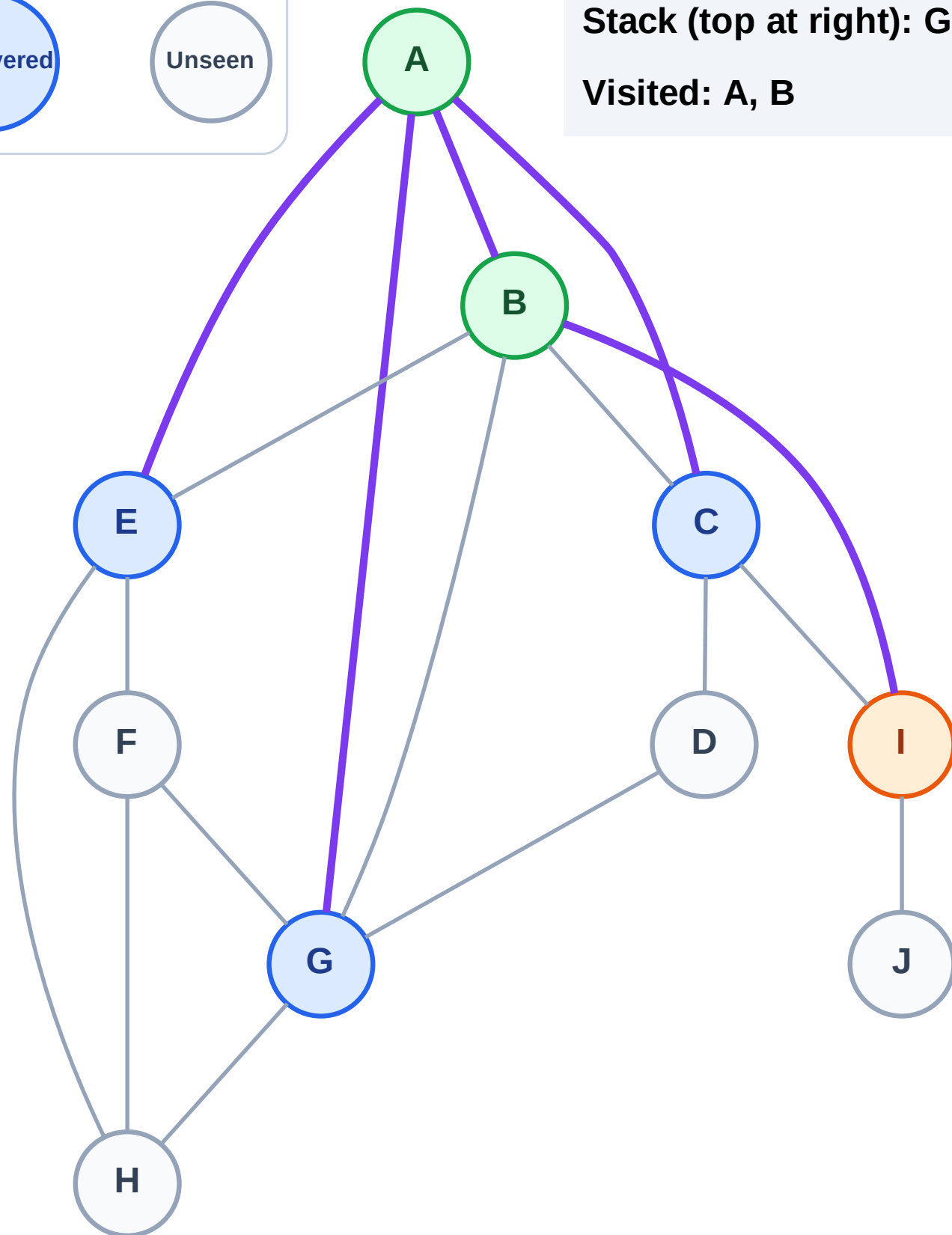
Step 6: Process node I

## Legend



Stack (top at right): G | E | C

Visited: A, B



# DFS Traversal

Step 7: Discover J from I

Legend

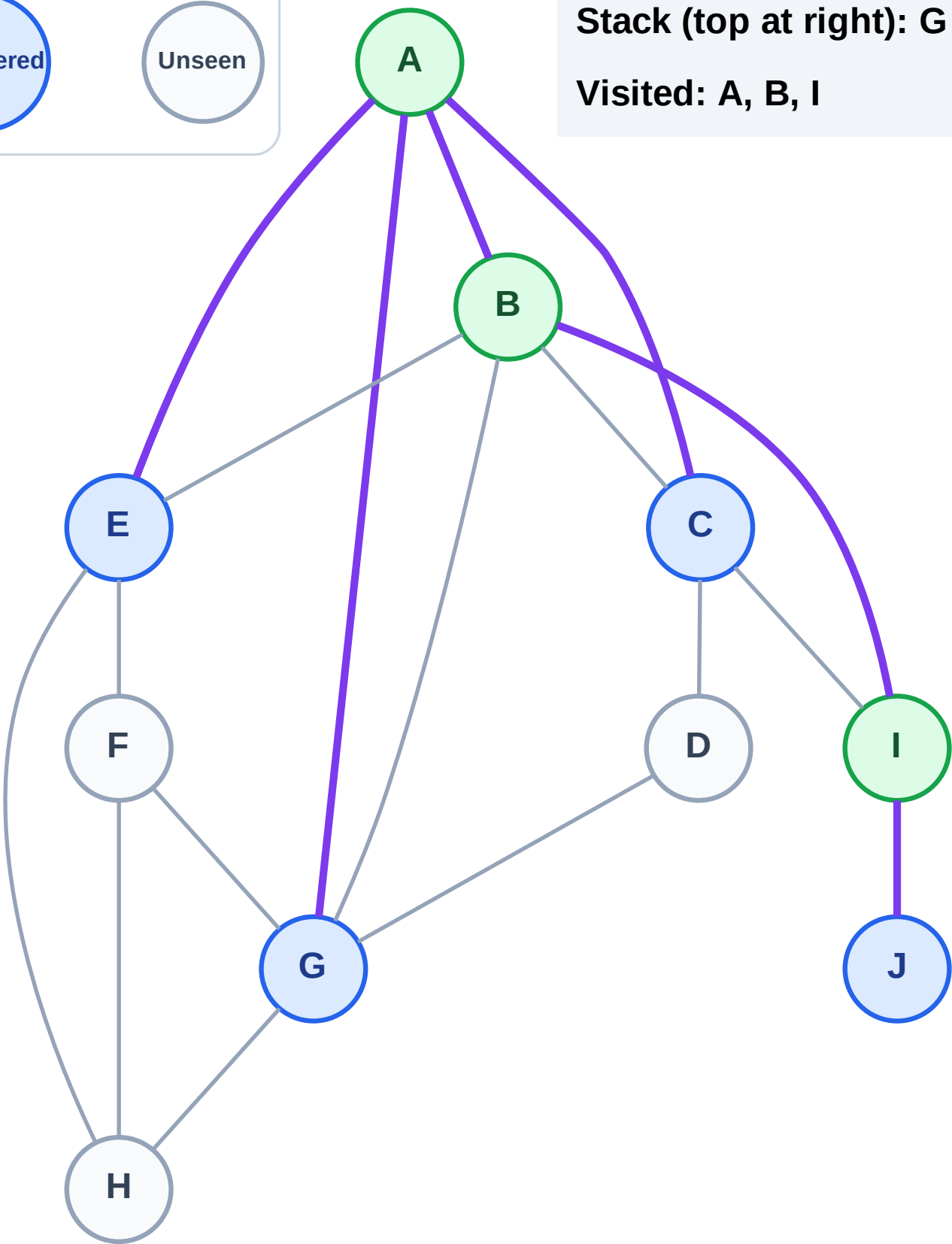
Complete

Processing

Discovered

Unseen

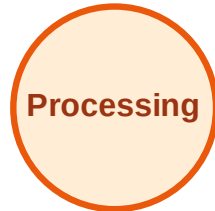
Stack (top at right): G | E | C | J  
Visited: A, B, I



# DFS Traversal

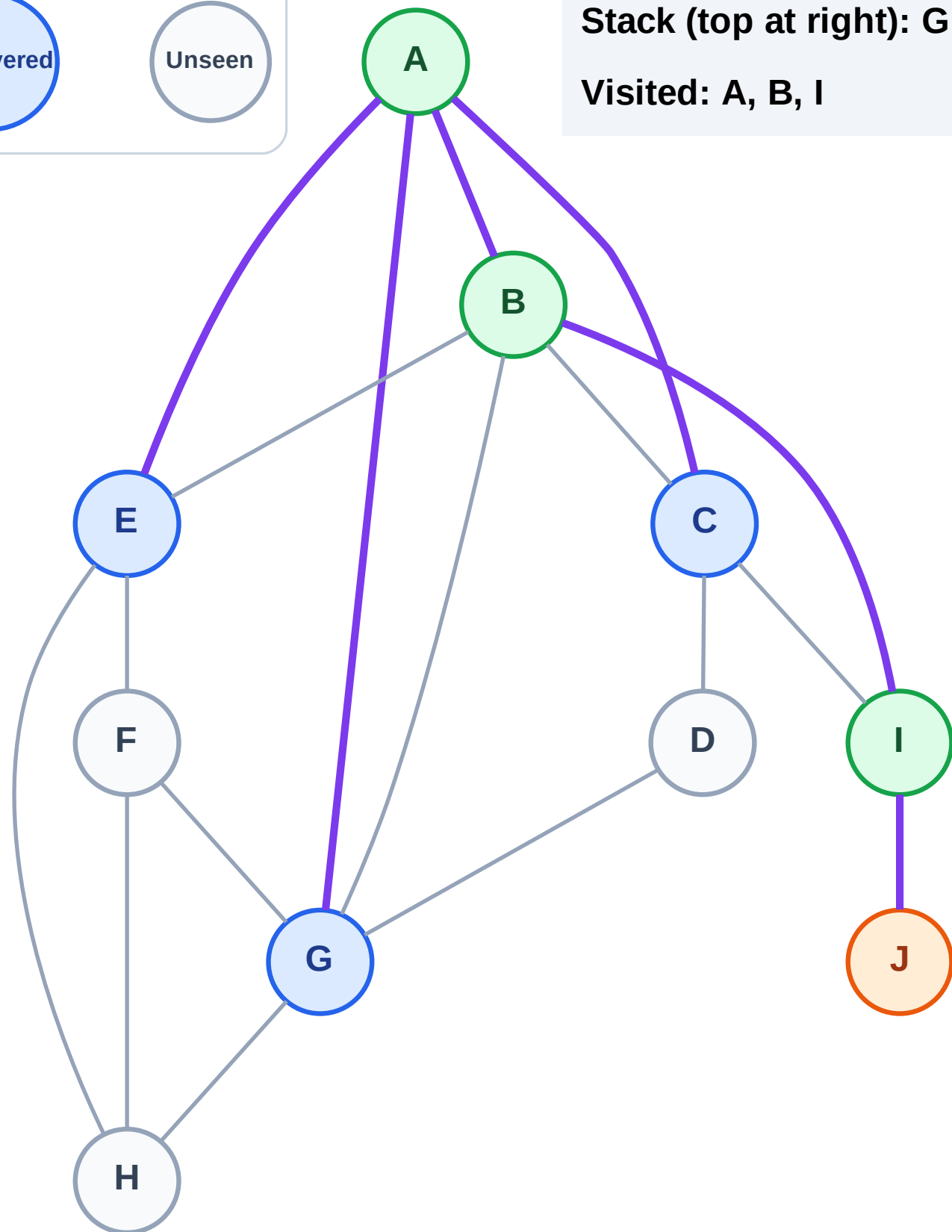
Step 8: Process node J

## Legend



Stack (top at right): G | E | C

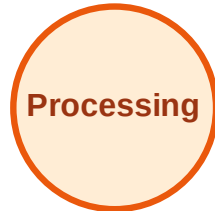
Visited: A, B, I



# DFS Traversal

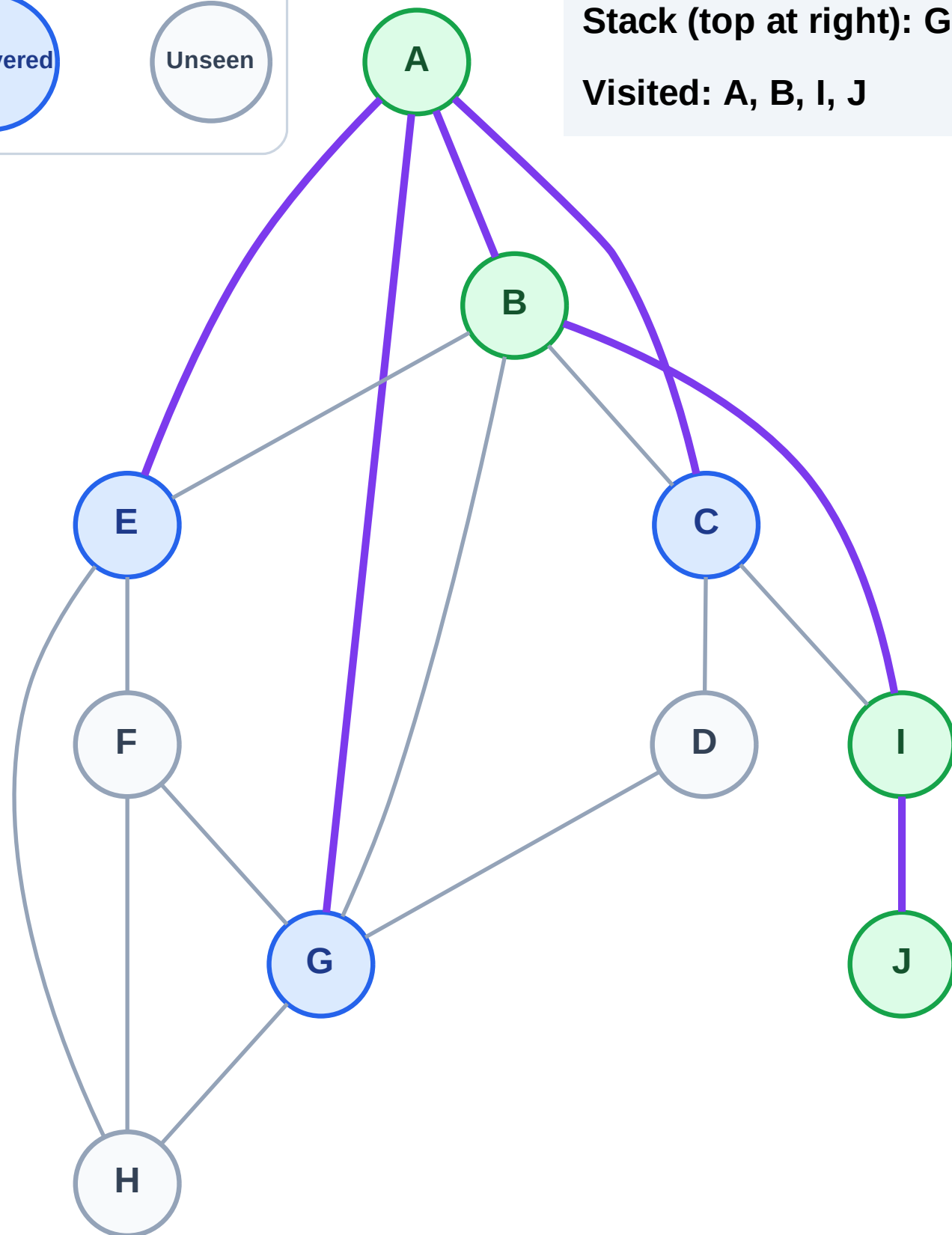
Step 9: No new neighbors from J

## Legend



Stack (top at right): G | E | C

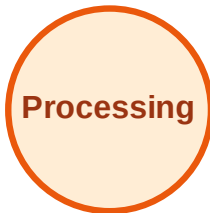
Visited: A, B, I, J



# DFS Traversal

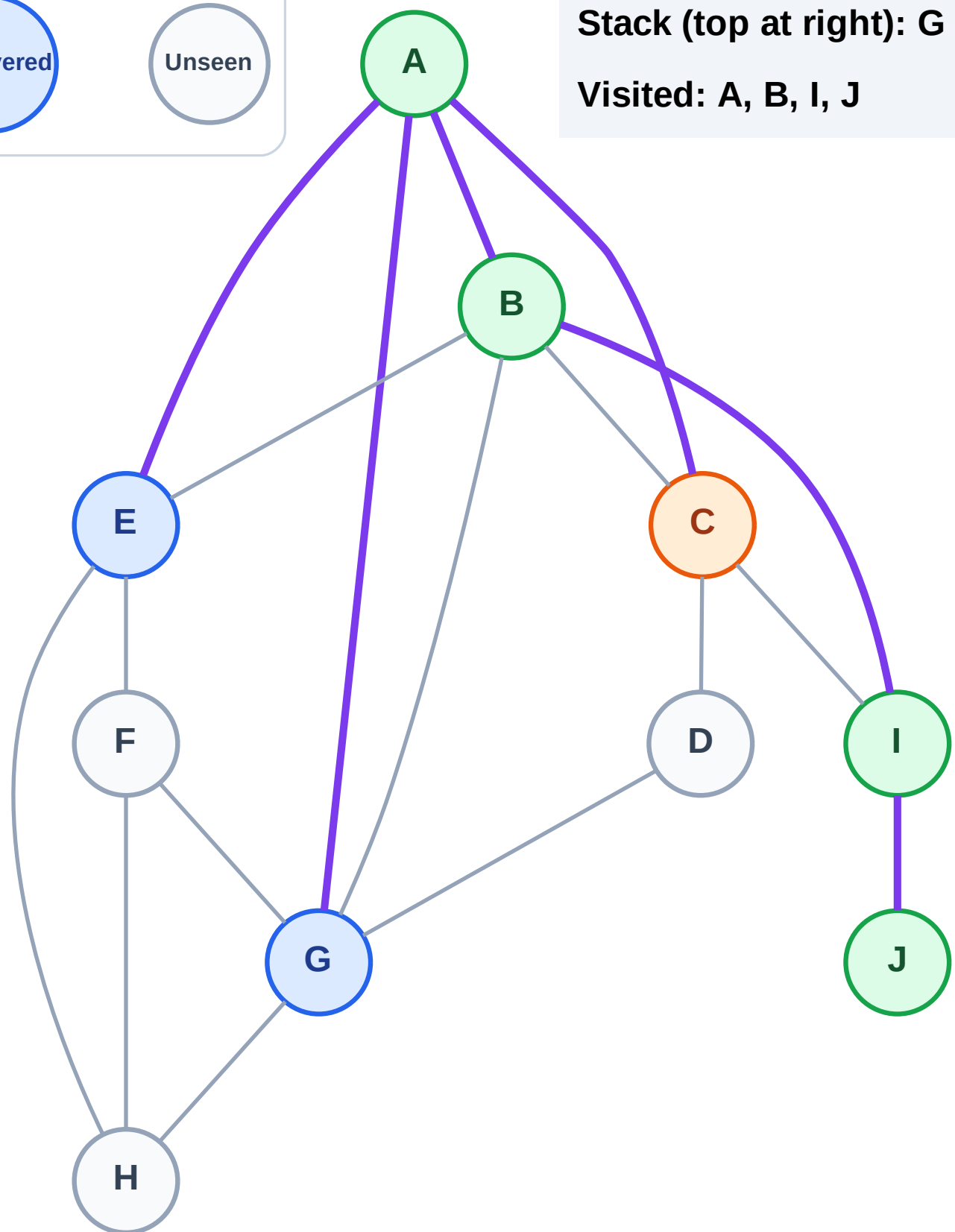
Step 10: Process node C

## Legend



Stack (top at right): G | E

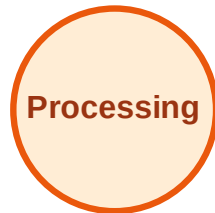
Visited: A, B, I, J



# DFS Traversal

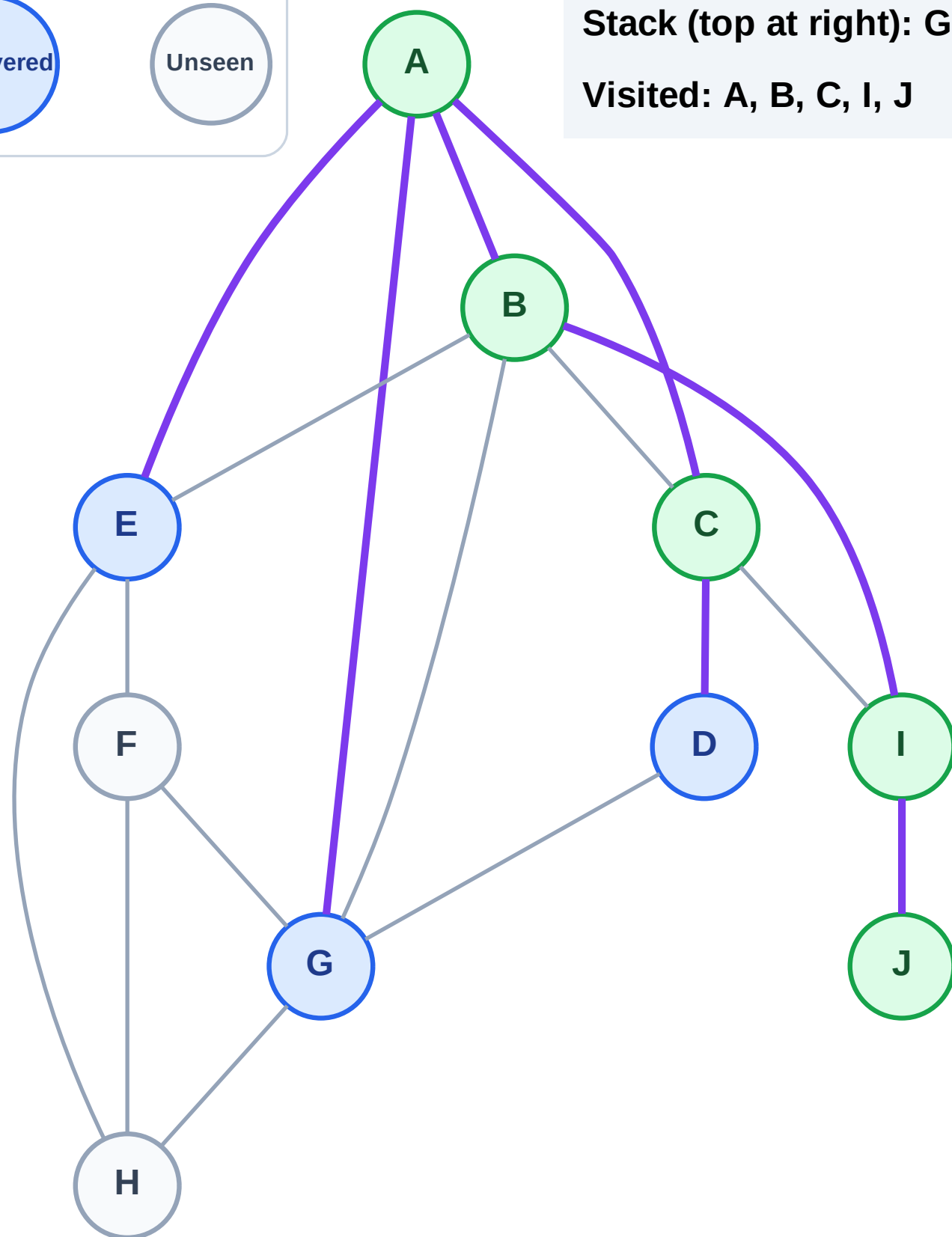
Step 11: Discover D from C

## Legend



Stack (top at right): G | E | D

Visited: A, B, C, I, J



# DFS Traversal

Step 12: Process node D

Legend

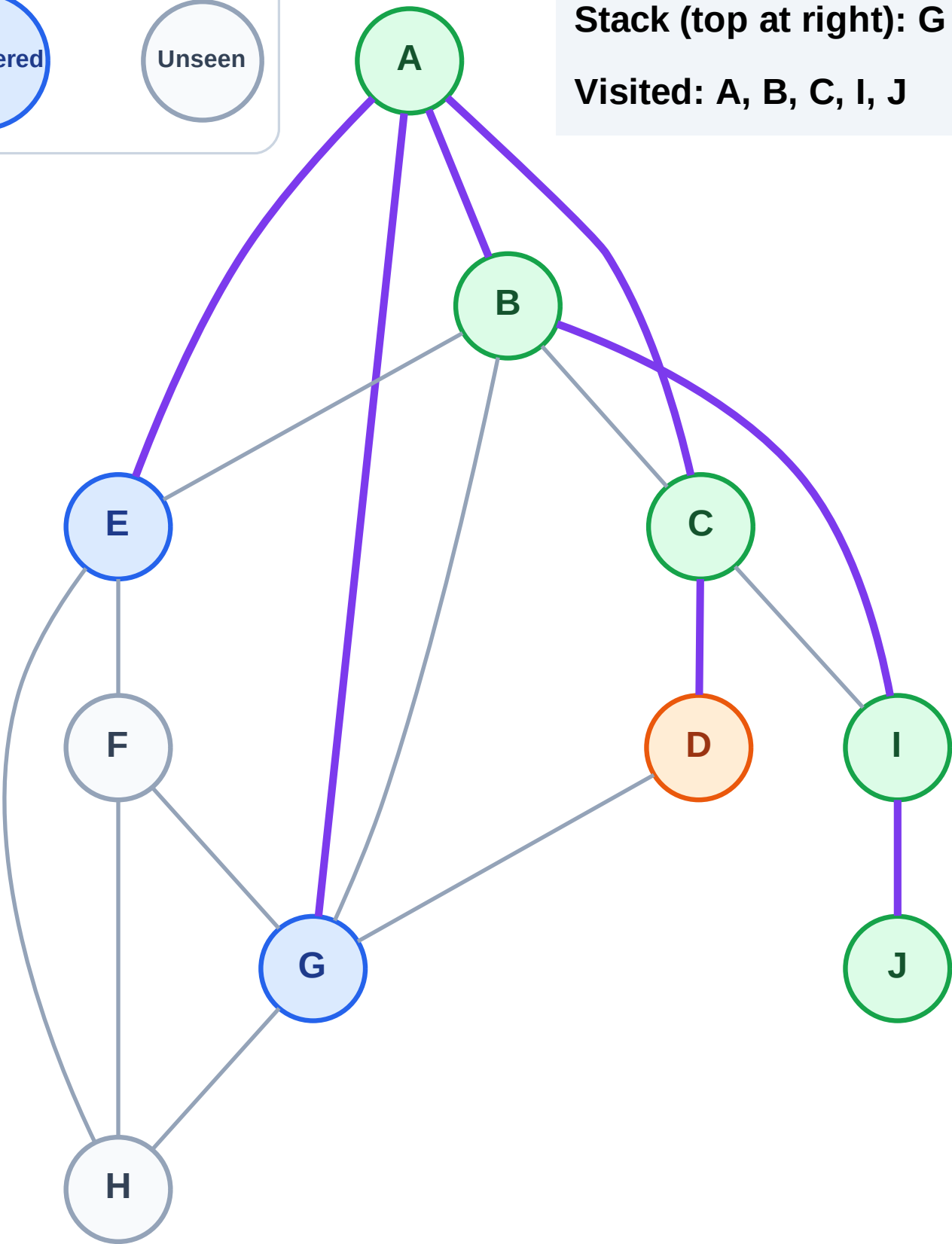
Complete

Processing

Discovered

Unseen

Stack (top at right): G | E  
Visited: A, B, C, I, J



# DFS Traversal

Step 13: No new neighbors from D

Legend

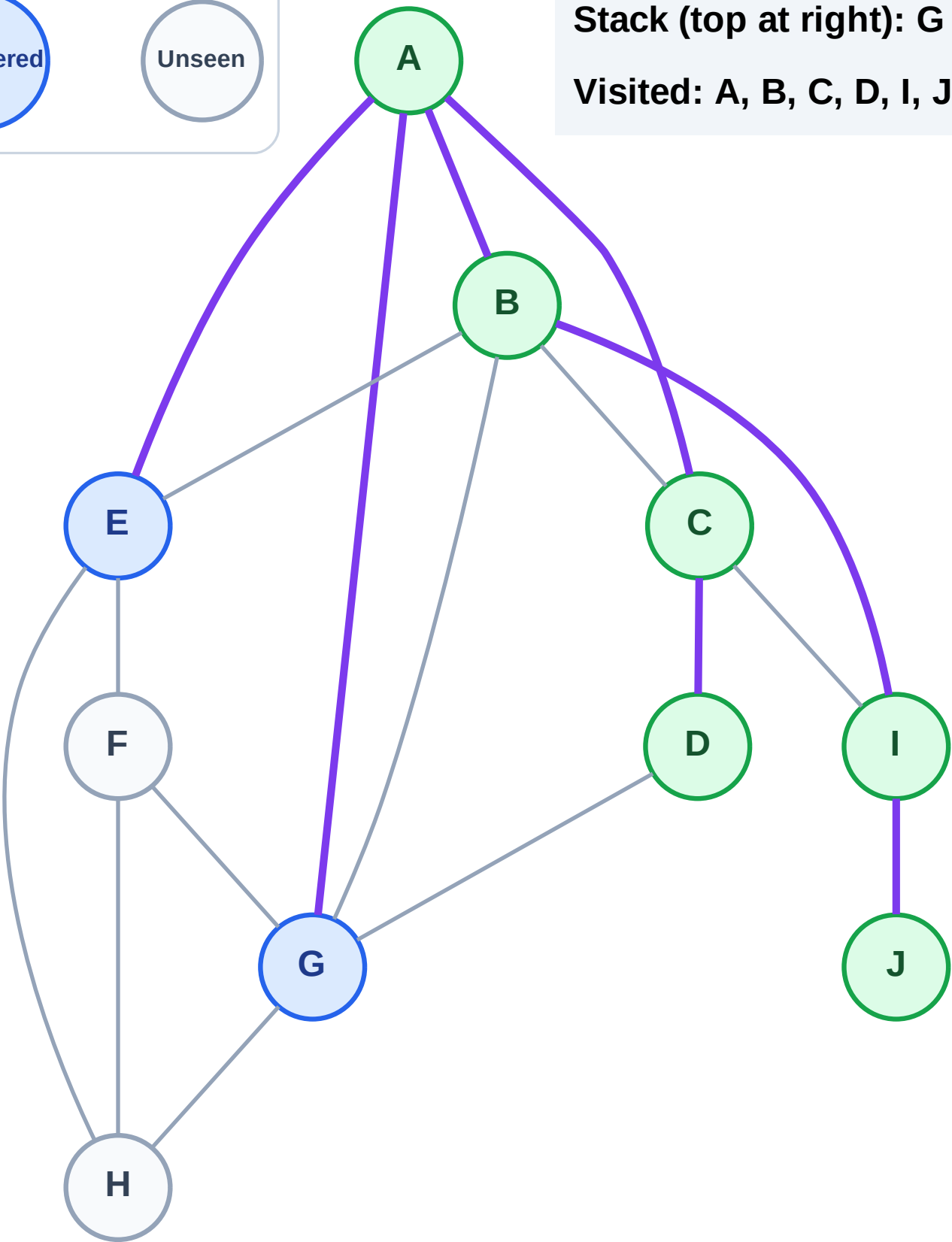
Complete

Processing

Discovered

Unseen

Stack (top at right): G | E  
Visited: A, B, C, D, I, J



# DFS Traversal

Step 14: Process node E

Legend

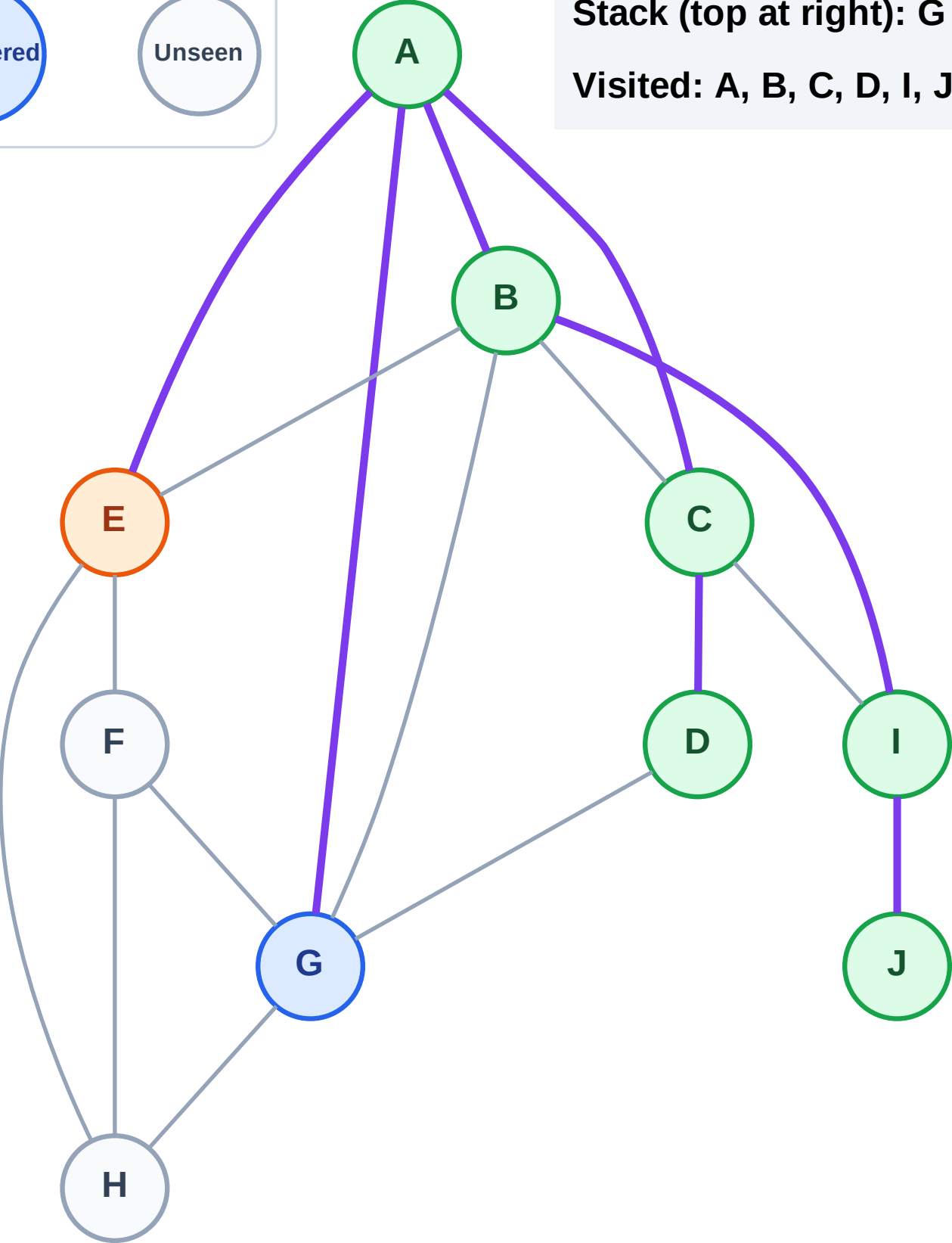
Complete

Processing

Discovered

Unseen

Stack (top at right): G  
Visited: A, B, C, D, I, J

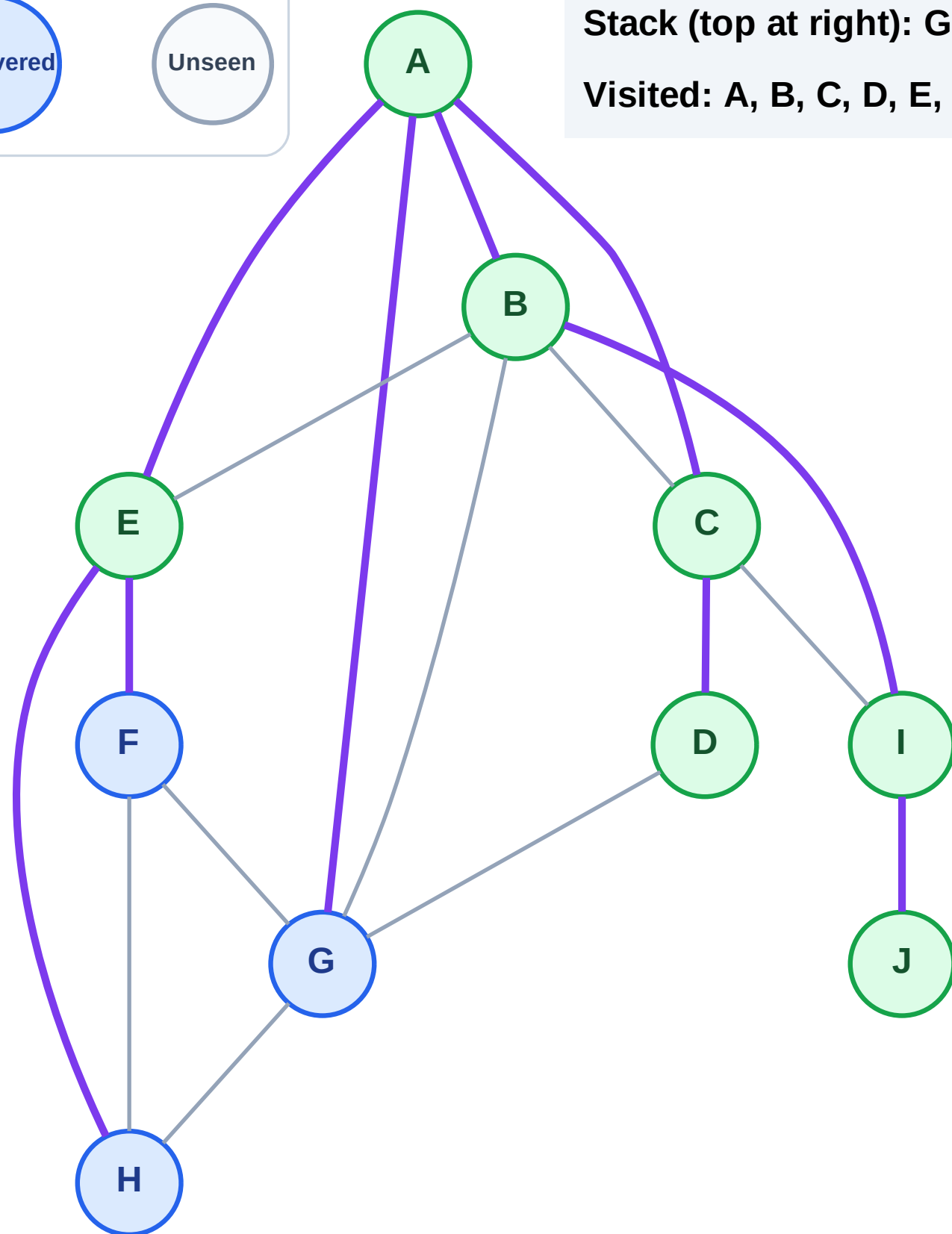


### Step 15: Discover H, F from E

### Legend

## Unseen

**Visited: A, B, C, D, E, I, J**



# DFS Traversal

Step 16: Process node F

Legend

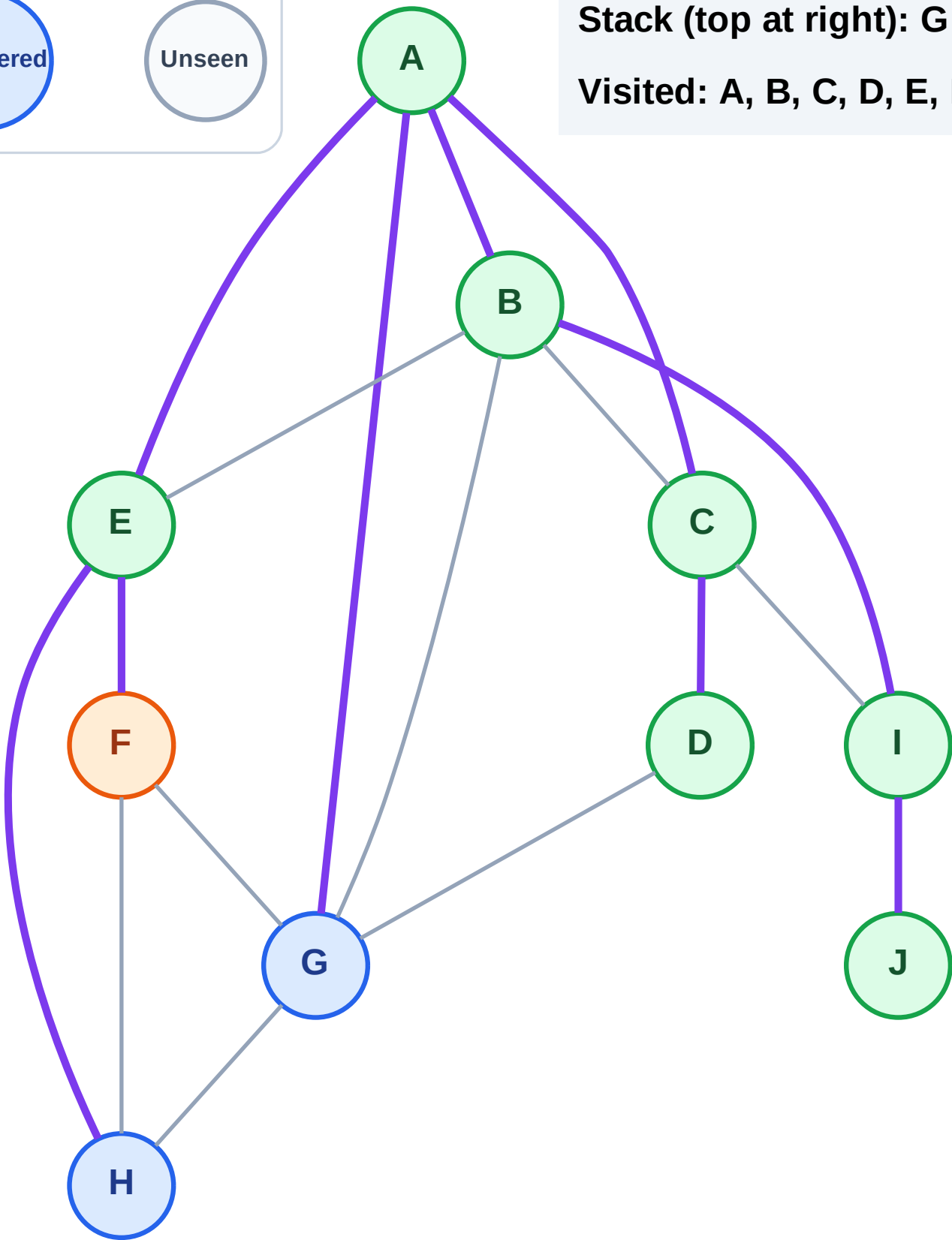
Complete

Processing

Discovered

Unseen

Stack (top at right): G | H  
Visited: A, B, C, D, E, I, J



# DFS Traversal

Step 17: No new neighbors from F

## Legend

Complete

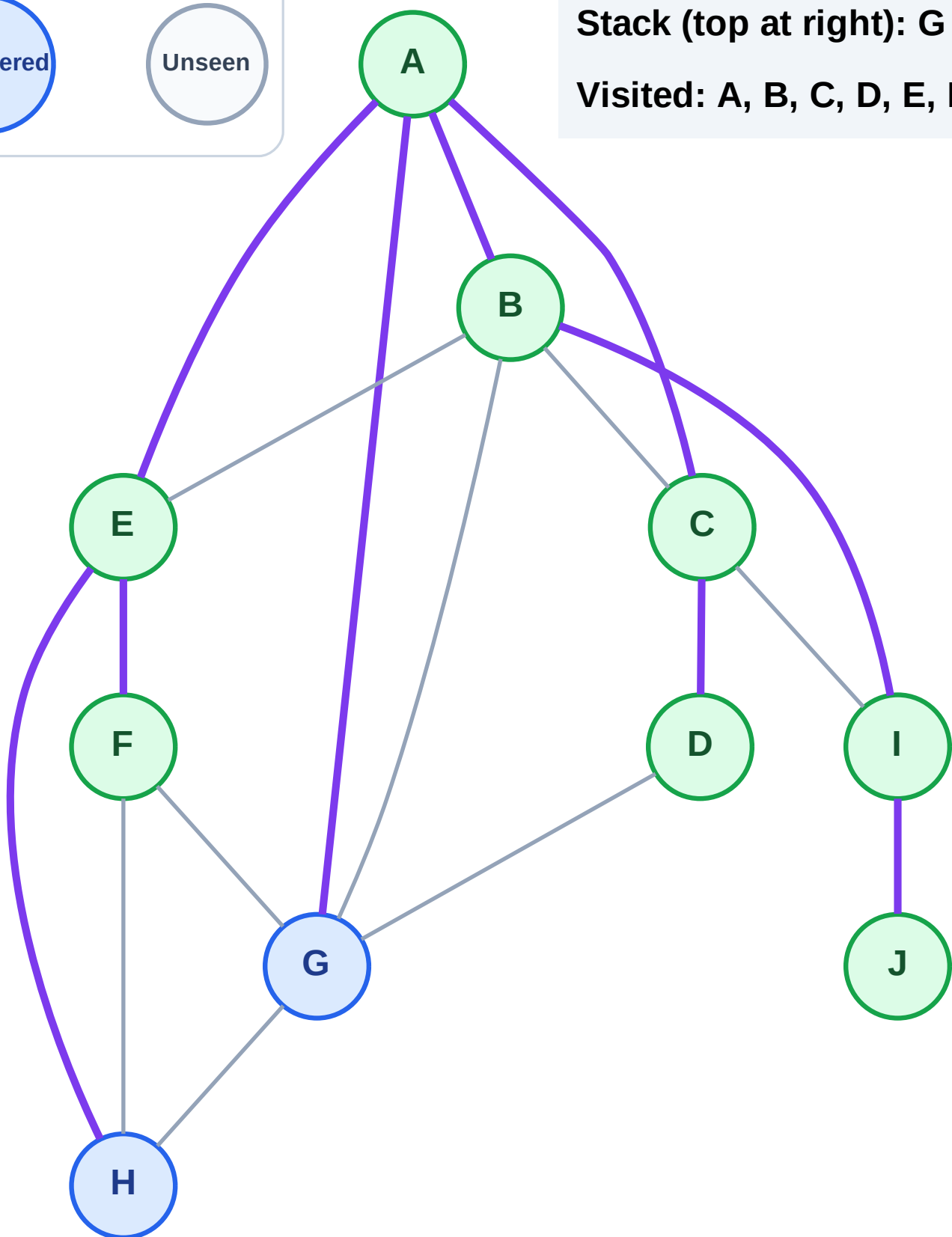
Processing

Discovered

Unseen

Stack (top at right): G | H

Visited: A, B, C, D, E, F, I, J



# DFS Traversal

Step 18: Process node H

Legend

Complete

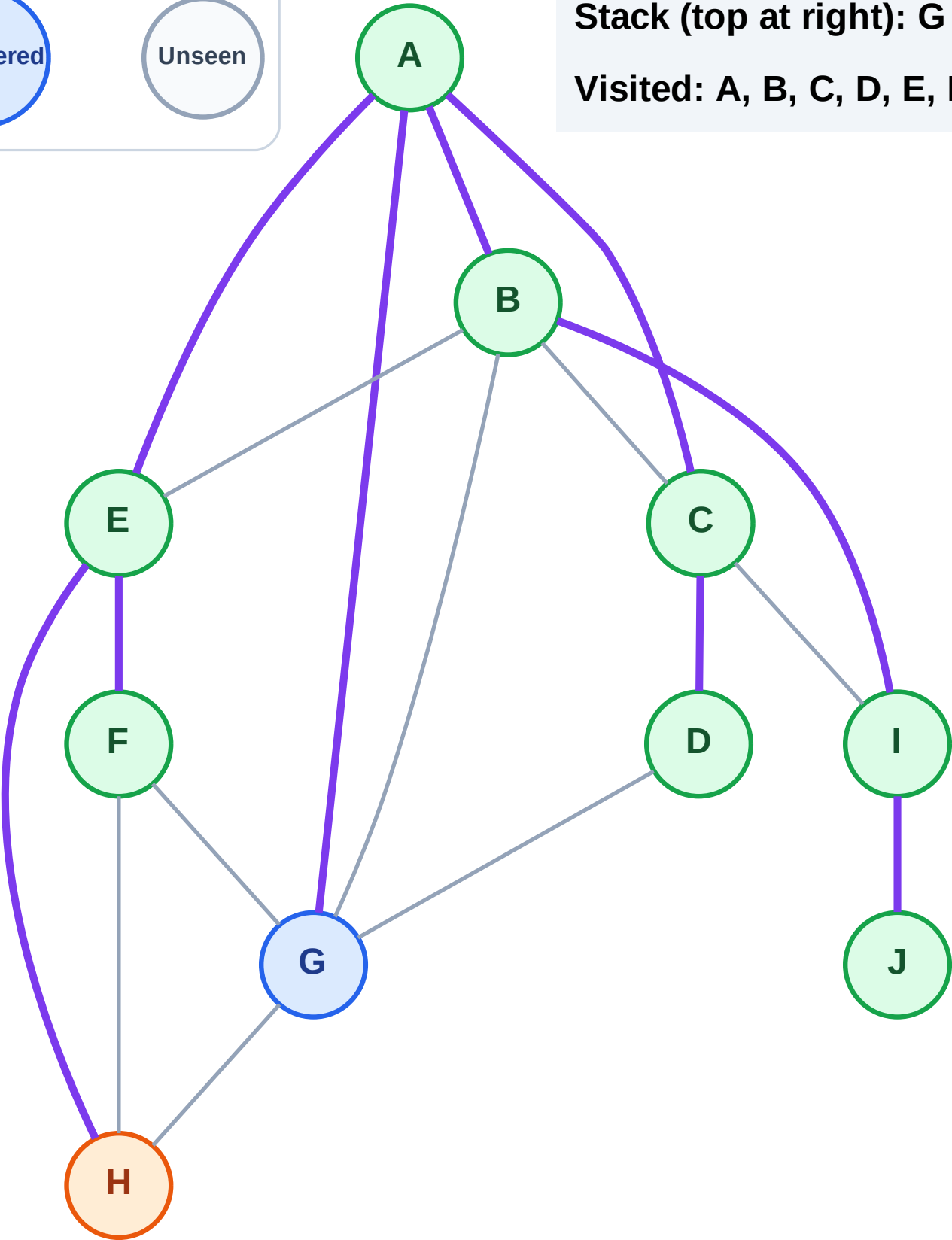
Processing

Discovered

Unseen

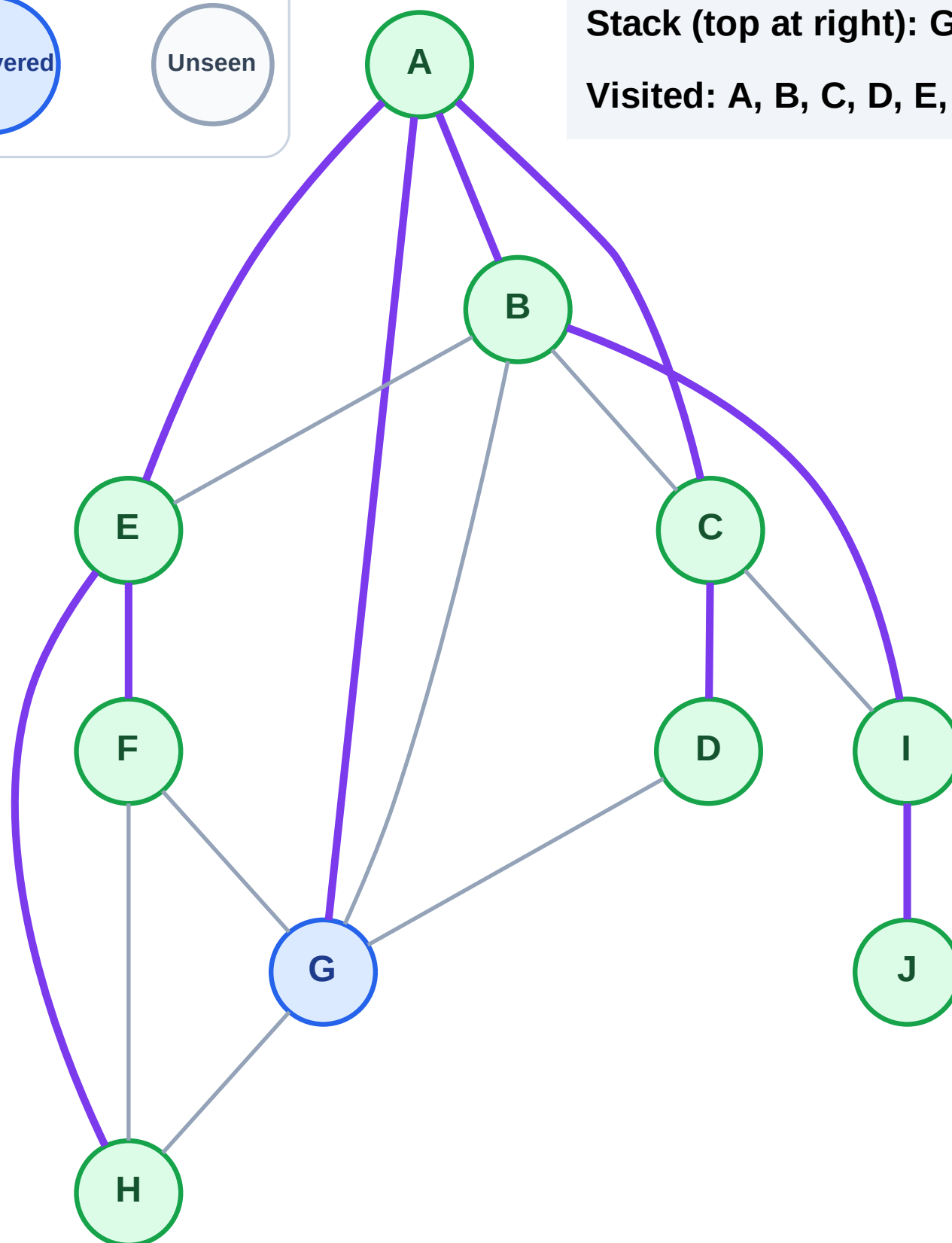
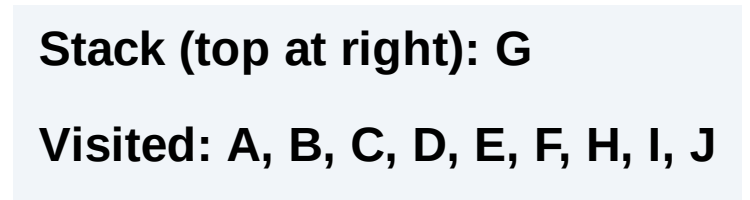
Stack (top at right): G

Visited: A, B, C, D, E, F, I, J



Step 19: No new neighbors from H

Step 19: No new neighbors from H



# DFS Traversal

Step 20: Process node G

Legend

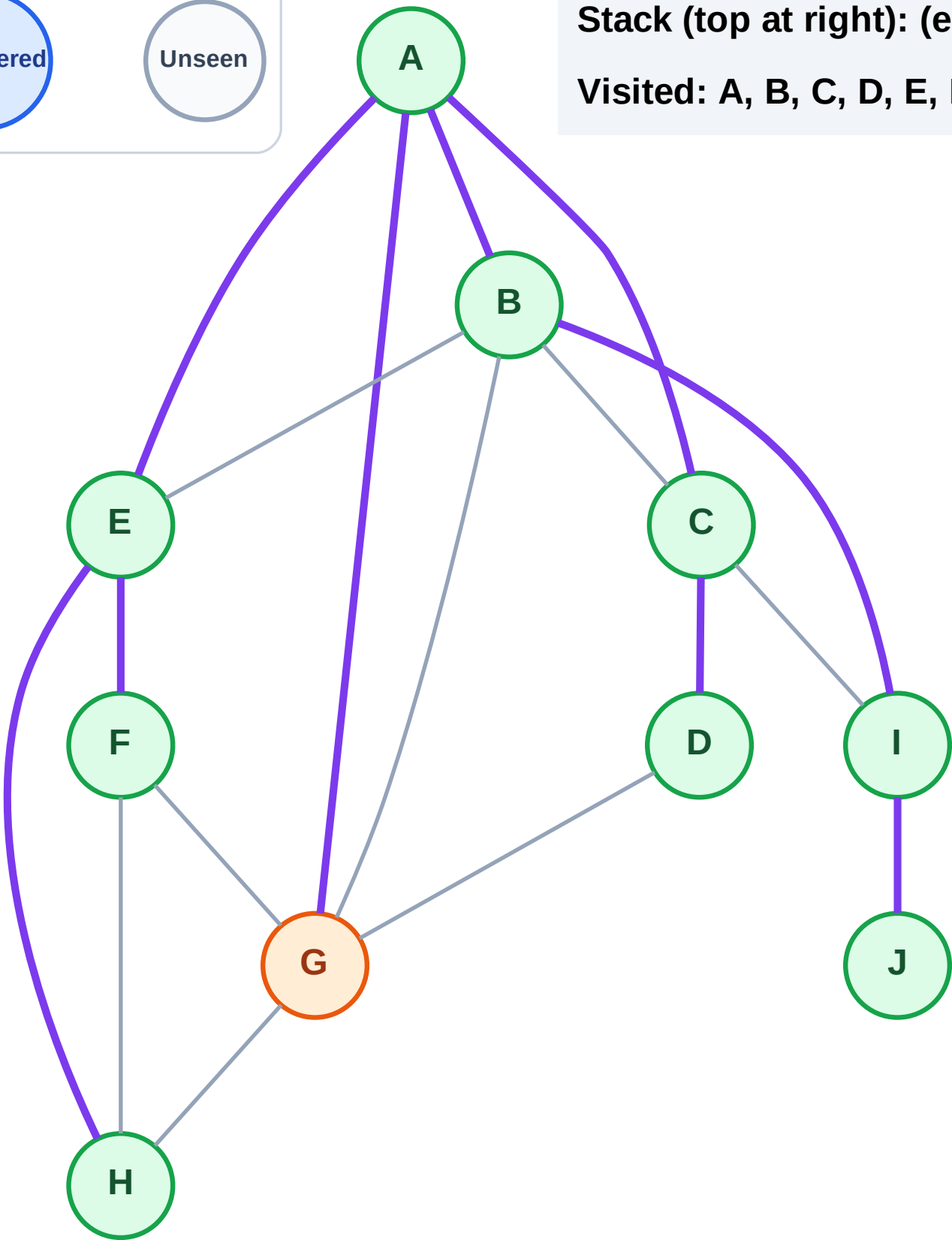
Complete

Processing

Discovered

Unseen

Stack (top at right): (empty)  
Visited: A, B, C, D, E, F, H, I, J



# DFS Traversal

Step 21: No new neighbors from G

## Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

